

# **F3 MIDTERM 3 EXAMS**

## **ALL SUBJECTS**

*Dear Students, Attempt These Midterm Exams!*

**For Marking Schemes Call 0705525657**

NAME: .....ADM NO ..... CLASS.....

**231**

**BIOLOGY**

**TIME: 2HRS**

### **INSTRUCTIONS TO CANDIDATES**

*:Answer ALL the questions .Answers should be written in the spaces provided*

1. a) Lietego school biology student used a microscope with X40 objective lens and X5 eye piece lens which had 2mm radius. Calculate the area of the field of view in micrometers.  
(2mks)

b) What is the average size of the cell in micrometers if there were 5 cells on the field of view (2mks)

2. State **three** differences between osmosis and active transport.  
(3mks)

| Osmosis | Active transport |
|---------|------------------|
|         |                  |
|         |                  |
|         |                  |

3. The lungs and ileum are adapted for absorption .State **three** features they have in common which facilitate absorption.

(3mks)

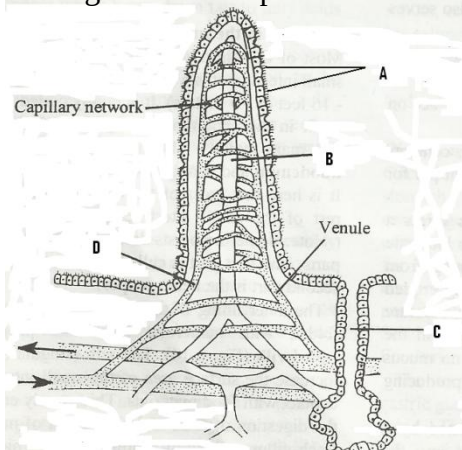
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4. The figure below represents a structure obtained from the ileum of a mammal.



a) Give the identity of the structure. (1mk)

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b) What is the importance of the structure named in (a) above? (1 mk)

.....

c) Name the parts labeled A, B and D. (3mks)

A..... B.....

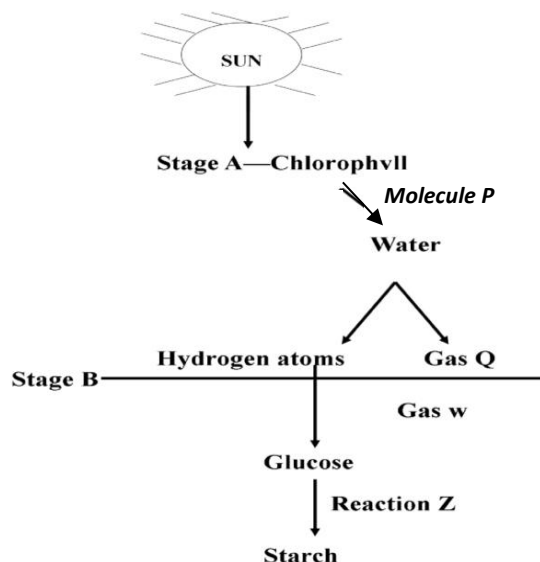
D.....

d) i) Name the juice secreted by the part labeled C.....(1mk)

ii) List **two** enzymes present in the juice named in d (i) above.  
(2mks) .....

.....

5. Below is a diagrammatic summary of the main biochemical events in photosynthesis. Study it carefully and answer the questions that follow.



a) Suggest the identify of molecule P. (1mk)

.....

b) Name the gases represented by the letters (2mks)

Q.....

W.....

c) Name the specific site for the reactions in stage B (1mk)

.....

d) Name reaction Z. (1mk)

Z .....

6. (a) What is the importance of the following substances in blood clotting process in man?

(i) Vitamin K:

.....(1mk)

(ii) Thrombin:

..... (1mk)

(b) Comment on:

(i) Disadvantage of having blood group O.

(1mk)

.....

.....

(ii) Advantage of having blood group AB.

(1mk)

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7. (a) State the reasons for the following adaptations of the xylem vessels.

(2mks)

(i) Narrow

lumen:

.....

(ii) Lack of cross walls:

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(b) State **two** distinguishing features of the phloem sieve tubes.

(2mks)

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8. (a) State **two** functional differences between arteries and veins in mammals.

(2mks)

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.....

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.....

9. a) List down **three** respiratory diseases in human beings

(3mks)

(i)

.....(ii).....(iii).....

.....

b) State the **three** theories that explain the mechanisms of opening and closing the stomata

(3mks)

.....

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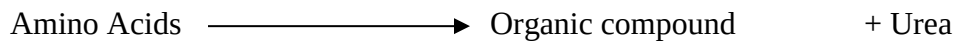
10. Name the nitrogenous wastes excreted by a fresh water fish

(1mk)

11. State the function of septum in the heart

(1mk)

12. The equation below represents a metabolic process that occurs in the mammalian liver.



(a) Name the

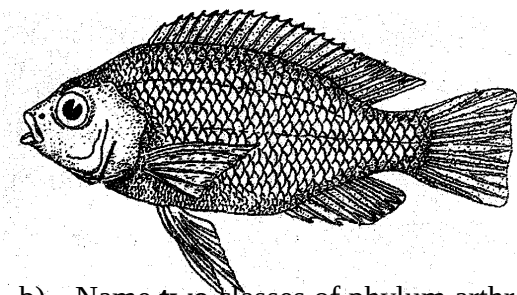
process.....(1m

k)

(b) What is the importance of the process to the mammal?

(2mks)

13. Below is a diagram of an organism



i) State the class which the organism belongs to (1mk)

ii) State two observable characteristics used to classify the organism in the class you have mentioned in (i) above (2mks)

b) Name **two** classes of phylum arthropoda with cephalothorax.

(2mks)

i) .....

.....

ii) .....

.....

14. In mitosis in animals chromatids failed to separate and move to opposite poles

a) Name the organelle that the cell was lacking

.(1mk)

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.....

b) Name **two** regions in plants where cells actively undergo mitosis

(2mks)

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15. a) What is the meaning of the following terms

(2mks)

i)

Autecology.....

...

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ii)

Synecology.....

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b) The number and distribution of stomata on three different leaves are shown in the table below.

| Leaf | Number of stomata |                 |
|------|-------------------|-----------------|
|      | Upper epidermis   | Lower epidermis |
| A    | 450               | 0               |
| B    | 185               | 270             |
| C    | 03                | 15              |

Suggest the possible habitats of the plants from which the leaves were obtained

(3mks)

A.....

...

B.....

.....

C.....

.....

- c) Apart from predation, state **two** other biotic factors that will influence the distribution of an organism in an ecosystem.

(2mks)

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16. What is the function of the following structures in the human reproductive organs?

- a) Prostate gland

(1mk)

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.....

- b) Uterus

(1mk)

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- c) Epididymis

(1mk)

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17. Define the term **oxygen debt** as used in biology

(1mks)

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18. a) Differentiate between incomplete and complete metamorphosis. Give example in each case.

.... (4mks)

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19. State **two** adaptations of a seed to its functions.

(2mk)

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20. Name the causative agent of cholera

(1mk)



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.....  
21. Define the term resolution power as used in microscopy  
(1mk)

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.....  
22. What is haemolysis  
(1mk)

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.....  
23. a) Name the compound that stores energy released during oxidation of glucose,  
(1mk)

.....  
.....  
b) A goat weighing 15.2kg requires 216KJ while a rat weighing 50g requires 2736KJ per day. Explain.  
(2mks)

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.....  
24. Explain why plants do not require complex excretory organs.  
(3mks)

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25. Name the instrument used in measuring wind velocity

(1mk)

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26. What is ecdysis

(1mk)

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# FORM 3 MIDTERM 3 EXAMS

NAME: ..... ADM NO .....

CLASS.....

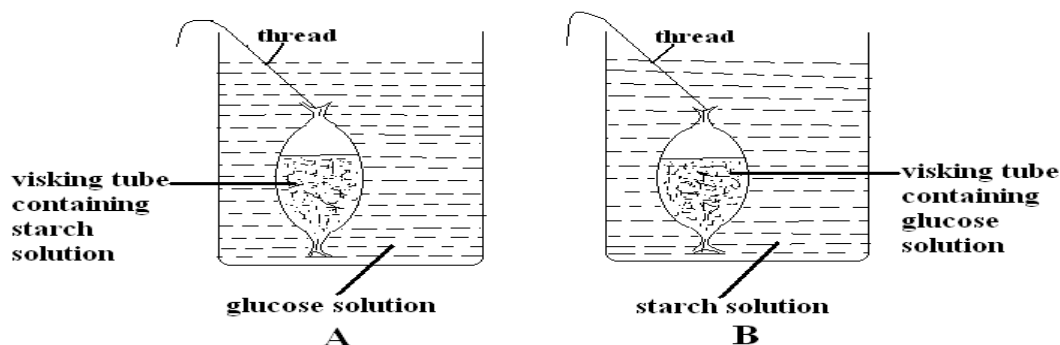
## 231/2 BIOLOGY

TIME: 2 HRS

### INSTRUCTIONS TO CANDIDATES:

- Answer **ALL** the questions
- Answers should be written in the spaces provided

1. The following experiment was set up by a form one class. After an hour, the contents of the visking tubing and the beaker were tested using iodine solution and benedict's solution.



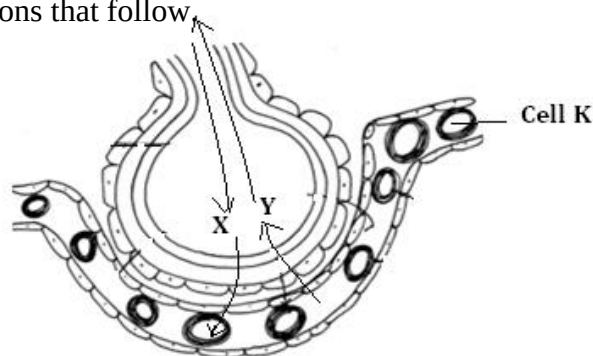
Record in the table below the expected observations after the contents in set up A and B were tested using iodine solution and benedict's solution.

[8 marks]

|        | Visking tubing  |                    | Beaker          |                     |
|--------|-----------------|--------------------|-----------------|---------------------|
| Set up | Iodine solution | Benedicts solution | Iodine solution | Benedict's solution |

|   |  |  |  |  |
|---|--|--|--|--|
| A |  |  |  |  |
| B |  |  |  |  |

2. The diagram below shows the association between the alveolus and a blood capillary. Study it and answer the questions that follow



- a) State the physiological process by which the gas labeled X enters the cell labeled K.

(1mk)

.....  
.....

- b) Identify gases represented by letter X and Y

X.....

.....(1mk)

Y

.....  
(1mk)

c) Give **two** adaptations of cell K to its functions.

(2mks)

.....  
.....  
.....

d) State **three** characteristics of respiratory surfaces.

(3mks)

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3. a) Define the following terms as used in animal nutrition

i) Dentition

(1mk)

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.....

ii) Homodont and heterodont teeth

(2mks)

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b) State **two** functions of ileum

(2mks)

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(c) Explain the importance of the following in the process of photosynthesis;

(2mks)

(i) Chlorophyll

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(ii) Light

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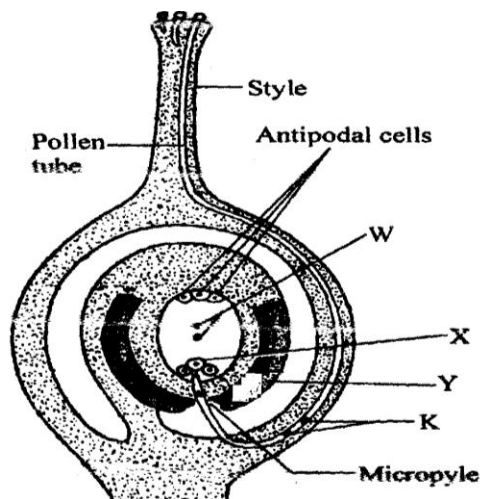
(d) State **one** use of Potassium in ( $K^+$ ) ion the body

(1mk)

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4. The diagram below shows a cross section through the female part of a flower.



a) Name the structures labelled W,X, and Y. (3mks)

X .....

Y .....

Z .....

b) State two functions of the pollen tube. (2mks)

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.....

c) What happens to antipodal cells after fertilization.

(1mk)

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d) Name the structure labelled K and state their role.

(2mks)

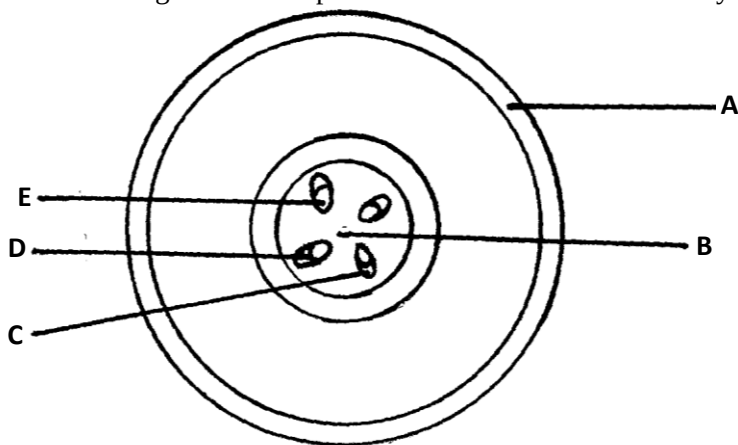
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5. The figure below represents a transverse section of a young stem.



a) Name the parts labelled **A** and **B** on the diagram. (2mks)

**A**.....

**B**.....

b) State the functions of the parts labelled **C**, **D** and **E**. (3mks)

**C**.....

**D**.....

**E**.....

c) List **three** differences between the section shown above and one of the same plant.

(3mks)

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SECTION B( 40 MARKS)

*Answer questions 6 ( compulsory)and either questions 7 or 8 in the spaces provided questions 8*

6. The glucose level in mg per  $100\text{cm}^3$  of blood was determined in two person Y and Z. Both had stayed for six hours without taking food. They were fed on equal amount of glucose at the start of the experiment .The amount of glucose in their blood was determined at intervals .The results are shown in the table below.

| Times in minutes | Glucose level in blood in mg / $100\text{cm}^3$ | Glucose level in blood in mg / $100\text{cm}^3$ |
|------------------|-------------------------------------------------|-------------------------------------------------|
|                  | Y                                               | Z                                               |
| 0                | 85                                              | 78                                              |
| 20               | 105                                             | 110                                             |
| 30               | 105                                             | 110                                             |
| 45               | 130                                             | 170                                             |
| 60               | 100                                             | 195                                             |
| 80               | 93                                              | 190                                             |
| 100              | 90                                              | 140                                             |
| 120              | 90                                              | 130                                             |
| 140              | 88                                              | 120                                             |

- a) On the grid provided, plot graphs of glucose levels in blood against time on the same axes. (7mks)

- b) What was the concentration of glucose in the blood of Y and Z at the 50<sup>th</sup> minute? (2mks)

Y

.....

.....

Z

.....

.....

- c) Account for the level of glucose in person Y



i) During the first 45 minutes.  
(2mks)

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ii) After 45<sup>th</sup> minute to the end of the experiment.  
(4mks)

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d) Account for the decrease in glucose level person Z after 60 minutes.  
(2mks)

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e) Low blood sugar level is harmful to the body. Explain.  
(3mks)

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- 7.a) State the various causes of seed dormancy .  
(8mks)
- b) Describe various factors that affect the process of seed germination.  
(12mks)
8. Discuss the adaptations of the mammalian skin to its functions.  
(20mks)

# FORM 3 MIDTERM 3 EXAMS

NAME..... ADM NO.....CLASS.....

**FORM THREE**

**CHEMISTRY**

**PAPER**

**1**

CODE:

| MAXIMUM<br>SCORE | STUDENTS<br>SCORE |
|------------------|-------------------|
| <b>80</b>        |                   |

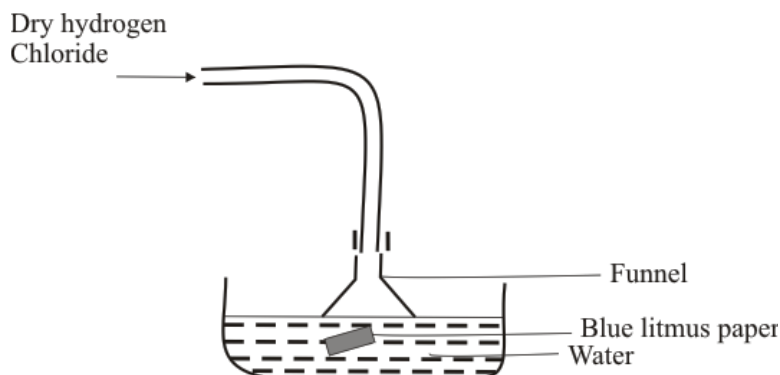
233/1

**TIME 2 HOURS**

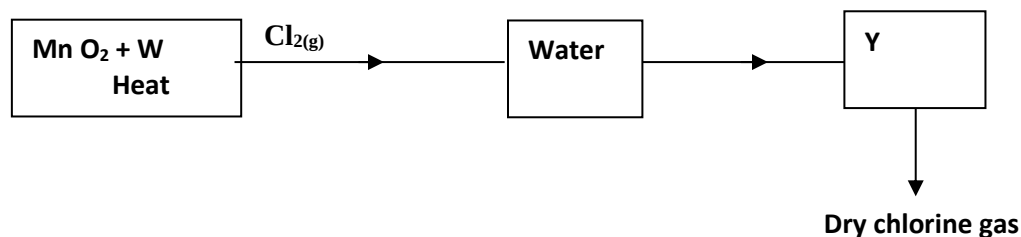
## INSTRUCTIONS TO CANDIDATES

- Write your name, class and Admission Number in the spaces provided above.
  - Answer **ALL** questions in the spaces provided.
  - Mathematical tables and electronic calculators may be used.
  - All workings **MUST** be clearly shown where necessary.
1. Explain why burning magnesium continues to burn in a gas jar full of sulphur (IV) oxide while a burning splint would be extinguished.  
(3 marks)
  2. Draw structural formulae and name two positional isomers with molecular formula C<sub>4</sub>H<sub>8</sub>.  
(2 marks)

3. Dry Hydrogen chloride gas was made to dissolve in water using the set of apparatus shown below



- (a) What is the use of the inverted funnel?  
(1 mark)
- (b) State and explain the observations made on the litmus paper  
(2 mark)
- (c) State and explain the observation made on the litmus paper if methylbenzene is used instead of water in the above set up.  
(2 mark)
4. Using sodium hydroxide solution, describe a chemical test that can be used to distinguish between copper (II) ions and iron (II) ions  
(2 marks)
5. The flow chart below shows laboratory preparation of chlorine gas. Study it and answer the questions that follow:



(a)  
Name

|   |   |  |  |  |  |   |   |  |  |
|---|---|--|--|--|--|---|---|--|--|
|   |   |  |  |  |  |   |   |  |  |
|   |   |  |  |  |  |   |   |  |  |
| I |   |  |  |  |  | Q | M |  |  |
|   | J |  |  |  |  |   | N |  |  |
| K | L |  |  |  |  | P |   |  |  |

substances (2 marks)

W-..... Y-.....

(b) What is the function of water in the above set up?  
(1 mark)

6. An unknown mass of anhydrous potassium carbonate was dissolved in water and the solution made up to 200cm<sup>3</sup>. 25cm<sup>3</sup> of this solution neutralized 18.0cm<sup>3</sup> of 0.22M nitric (v) acid. Calculate the unknown mass of potassium carbonate (**K**=39, **C**=12, **O**=1)  
(3 marks)

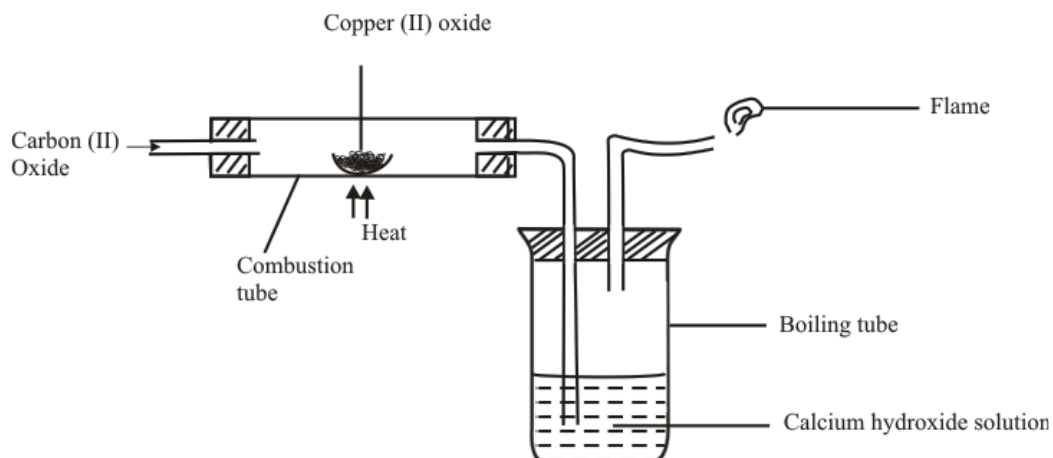
7. Below is a sample of the periodic table

a) Give the family name to which elements **M** and **N** belong  
(1 mark)

(b) Compare the reactivity of elements **I** and **K**. Give a reason  
(2 mark)

(c) Write the formula of the compound formed when **P** reacts with **Q**  
(1 mark)

8. Study the experimental set up of apparatus shown below.



(i) State two observations made in the set up as the experiment progressed  
(2 marks)

(ii) Using an equation, explain the change that occurred in the boiling tube  
(1 mark)

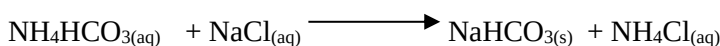
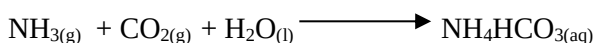
(iii) Why was the gas burned in the flame?  
(1 mark)

9. Painting, oiling, galvanizing and tin plating are methods of rust prevention.

(a) Explain the similarity of these methods in the way they prevent rusting  
(1 mark)

(b) Explain why galvanized iron objects are better protected even when scratched  
(1mark)

10. The chemical equations below are the main reactions in large scale manufacture of sodium carbonate.



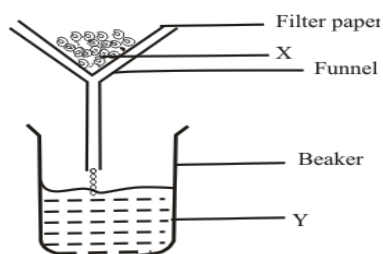
(a) Explain how the two products  $\text{NaHCO}_3$  and  $\text{NH}_4\text{Cl}$  are separated  
(1 mark)

(b) How sodium carbonate is finally obtained from  $\text{NaHCO}_3$ ?  
(1 mark)

(c) Explain how ammonia is recovered in this process.  
(1 mark)

11.  $80 \text{ cm}^3$  of oxygen gas diffused through a porous hole in 50 seconds. How long will it take  $120 \text{ cm}^3$  of Nitrogen (IV) oxide to diffuse through the same hole under the same conditions? (N = 14, O = 16) (3 marks)

12. Filtration is carried out in the apparatus shown



Name :  
marks)

(2

X ..... Y .....

13. Two carbonates **P** and **Q** are weighed before and after heating. The results are given in the table below.

| Carbonate | Mass in grams  |               |
|-----------|----------------|---------------|
|           | Before heating | After heating |
| P         | 15.0           | 15.0          |
| Q         | 15.0           | 10.0          |

Which one is likely to be sodium carbonate? Explain.  
(2 marks)

14. Describe how you would separate a solid mixture of lead(II) chloride and copper(II) oxide (3 marks)

15. The general formula for a homologous series of organic compounds in  $\text{C}_n\text{H}_{2n+2}$



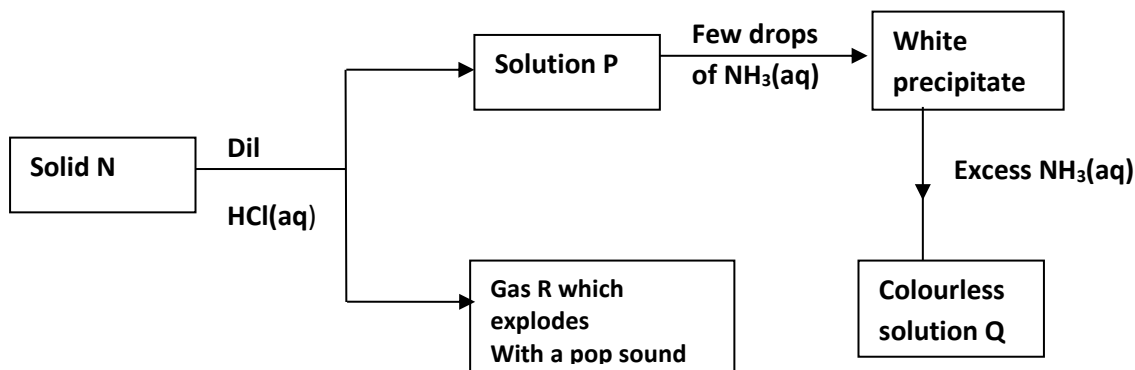
(a) Give the name and structural formula of the fourth member of the series  
(2 marks)

(i) Name .....

(ii) Structural formula

(b) Write an equation for the combustion reaction of the above molecule  
(1mark)

16. The scheme below shows some reactions sequence starting with solid N. Study it and answer the questions that follow:



(a) Identify solid N.....  
(1 mark)

b) Write the equation for the formation of the colourless solution Q  
(1 mark)

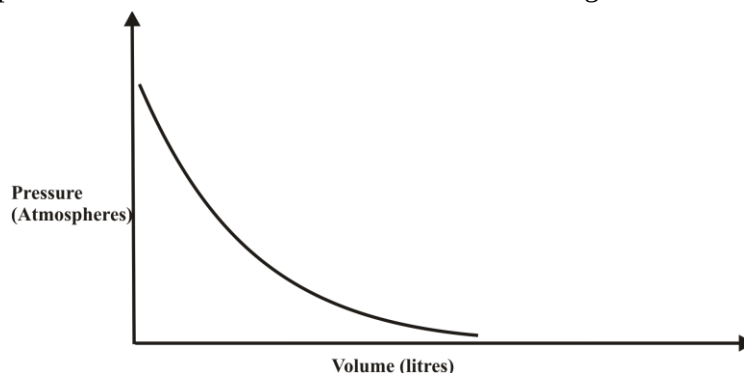
c) Give the identity of gas R  
(1 mark)

17. In an experiment, a gas jar containing moist sulphur (IV) oxide was inverted over another gas jar containing hydrogen sulphide gas.

(a) State and explain the observation that was made  
(2 marks)

(b) State the precautions that should be taken when carrying out this experiment  
(1 mark)

18. The graph below shows the behavior of a fixed mass of a gas at constant temperature



- (a) What is the relationship between the volume and pressure of the gas?  
(1 mark)

- (b) 3 litres of oxygen gas at one atmosphere pressure were compressed to two atmospheres at constant temperature. Calculate the new volume occupied by the oxygen gas.  
(2 marks)

19. The table below shows the relative atomic masses and percentages abundance of the isotopes  $M_1$  and  $M_2$  of element  $M$

|       | Relative abundance | % abundance |
|-------|--------------------|-------------|
| $M_1$ | 60.57              | 59.71       |
| $M_2$ | 62.83              | 40.29       |

- Calculate the relative atomic mass of element  $M$   
(2 marks)

20. The table below shows the pH values of solutions **A**, **B**, **C** and **D**

|          |   |   |   |   |
|----------|---|---|---|---|
| solution | A | B | C | D |
|----------|---|---|---|---|

|    |   |   |    |    |
|----|---|---|----|----|
| pH | 2 | 7 | 11 | 14 |
|----|---|---|----|----|

- (a) Which solution is likely to be that of calcium hydroxide  
(1 mark)
- (b) Select the solution in which a sample of aluminum oxide is likely to dissolve. Give a reason for your answer

21. Name one property of neon that makes it possible to be used in electric lamps.  
(1mark)

22. Distinguish between ionic bond and covalent bond  
(2 marks)

23. Explain why the boiling point of hexane is higher than that of ethane. (relative molecular mass of ethane is **30** while that of hexane is **86**)  
(2 marks)

24. When a student was stung by a nettle plant, a teacher applied an aqueous solution of ammonia to the affected area of the skin and the student was relieved of pain. Explain  
(2 marks)

25. Using dots(.) and crosses (x) show the bonding  
(3mks)

(i)  $\text{NH}_4^+$

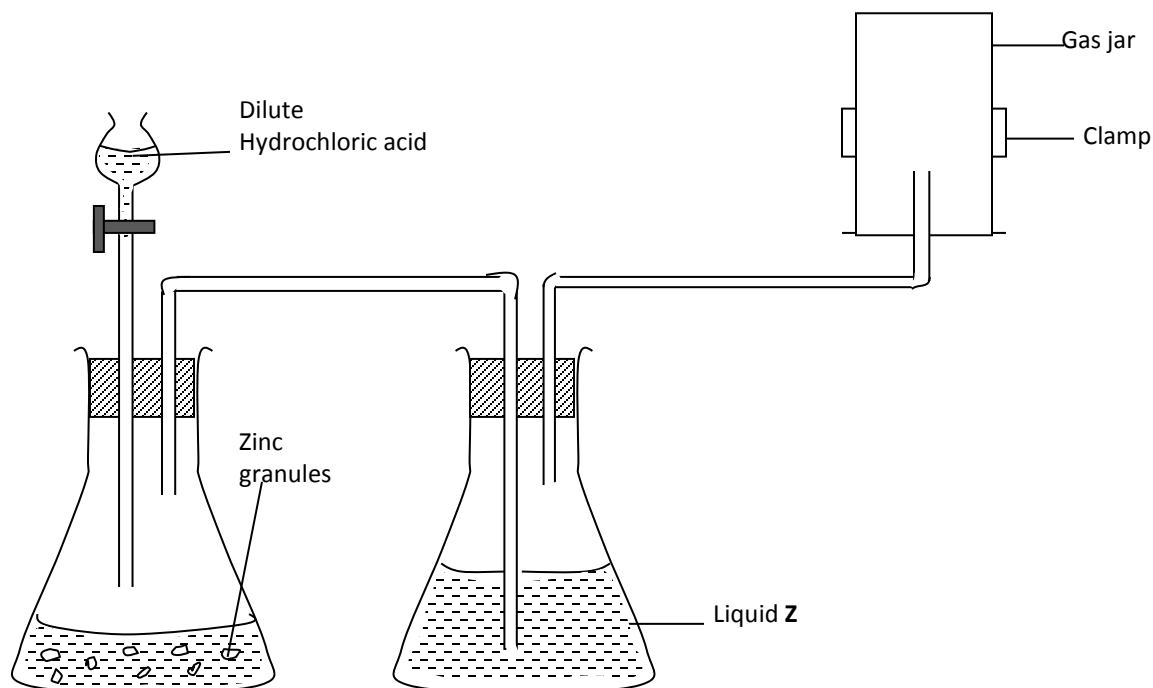
(ii) Carbon (IV) oxide.

(iii) Potassium chloride

26. On complete combustion of a hydrocarbon gas X, 1.32g of carbon (IV) oxide and 0.54g of water.  
Calculate the empirical formula of X (C = 12.0, H = 1, O = 16.0)  
(3 marks)

27. Study the diagram below and answer the questions that follow.

(3 marks)



- (a) Write an equation for the reaction between zinc granules and dilute hydrochloric acid.  
(1 mark)
- (b) What property of hydrogen is demonstrated by the method of collection shown on the diagram?  
(1 mark)
- (c) Hydrogen gas passed through liquid Z. What is the name of liquid Z and what is the purpose of liquid Z?  
(2 mark)
- (ci) Name one industrial use of hydrogen.  
(1 mark)

28. Three liquids were mixed together accidentally and this included lubricating oil, kerosene and water. The table below gives information about the properties of the liquids.

| Constituent     | Boiling point in $^{\circ}\text{C}$ | Solubility in water | Solubility kerosene |
|-----------------|-------------------------------------|---------------------|---------------------|
| Lubricating oil | 350 – 400                           | Insoluble           | Soluble             |
| Kerosene oil    | 175 – 250                           | Insoluble           |                     |
| Water           | 100                                 |                     | Insoluble           |

Suggest a method you would use to separate the three liquids.

(2 marks)

29. a) Define the term allotropy

(1mk)

b) Name the two allotropes of sulphur

(2 mks)

30. A concentrated solution of Sulphuric (VI) acid contains 70%  $\text{H}_2\text{SO}_4$  and has a density of  $1.8\text{g cm}^3$ .

Determine the molarity of Sulphuric (VI) acid solution. (H= 1, S=32, O=16)

(3 mrks)

# FORM 3 MIDTERM 3 EXAMS

NAME:.....CLASS:.....ADM

NO:.....

## CHEMISTRY

PAPER 2 233/2

TIME: 2 HOURS

### INSTRUCTIONS TO CANDIDATES

- ❖ Write your name, class and Admission Number in the spaces provided above.
- ❖ Answer **ALL** questions in the spaces provided
- ❖ Mathematical tables and electronic calculators **MAY** be used
- ❖ All workings **MUST** be clearly shown where necessary

1. The table below shows the information concerning elements S, T, U, V, and W. the letter are not the actual symbols of the elements. **Study** it and answer the questions that follow.

| Element | Period | Formula of oxide                   |
|---------|--------|------------------------------------|
| S       | 2      | S <sub>2</sub> O                   |
| T       | 3      | T <sub>2</sub> O <sub>3</sub>      |
| U       | 3      | UO <sub>2</sub> or UO <sub>3</sub> |
| V       | 3      | Does not form oxide                |
| W       | 4      | W <sub>2</sub> O                   |

- (a) Write down:-
- (i) The electronic arrangement of the element W.....  
(1mk)
  - (ii) The formula of the ion formed by element T.....  
(1mk)
- (b) Two of the oxides, S<sub>2</sub>O and UO<sub>3</sub> are apparently dissolved in distilled water. Compare the PH value of the resulting solutions.  
(2mks)
- (c) **Compare** with an explanation the following.

**FOR MARKING SCHEMES CALL/TEXT/WHATSAPP 0705525657**

(i) The reactivity of S with that of W.  
(2mks)

(ii) The electrical conductivity of T with that of magnesium.  
(2mks)

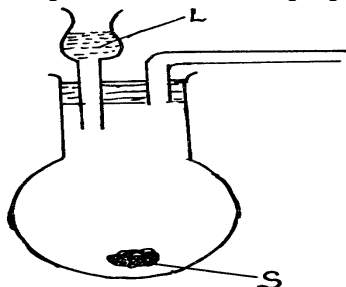
(iii) The melting point of U with that of X which is just below U in the group.  
(2mks)

(d) **Write** the electronic configuration of V. ....  
(1mk)

(e) **Select two** elements which are non – metals.  
(1mk)

(f) **Select two** elements which belong to the same group.  
(1mk)

2. The set up below is used to prepare and collect dry samples of hydrogen sulphide gas.



a) Name suitable substances for use as  
(2mks)

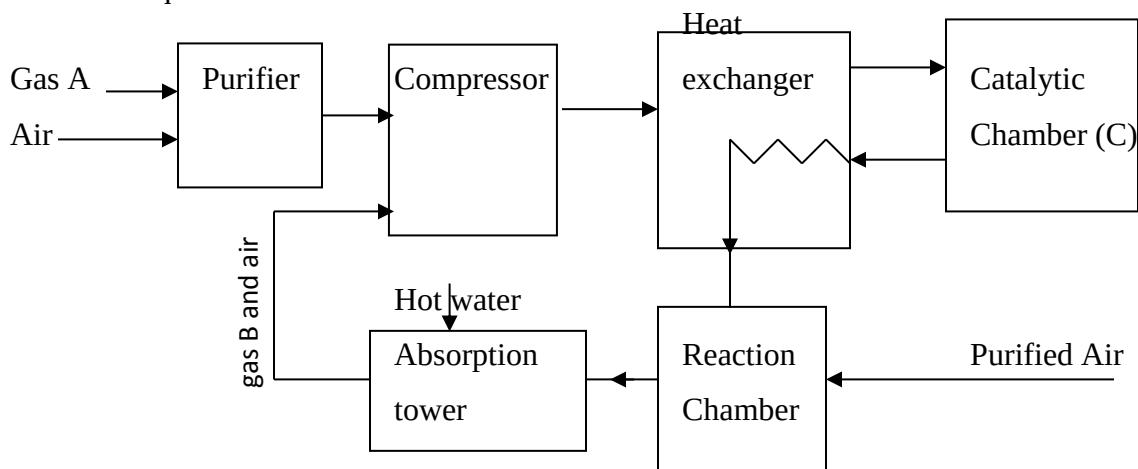


- (i) L .....
- (ii) S .....
- b) Complete the diagram to show how dry hydrogen sulphide gas is obtained and collected.  
(3 mks)
- c) Write a balanced equation for the reaction between L and S named in (a) above.  
(1mk)
- d) (i) State the effect of hydrogen sulphide gas on litmus.  
(1mk)
- (ii) State a chemical test for hydrogen sulphide gas.  
(1mk)
- (iii) What do you observe when hydrogen sulphide gas is passed through aqueous zinc chloride  
(1mk)
- e) (i) Name the process used to extract sulphur from the ground in Louisiana and Texas.  
(½mk)
- (ii) State the uses of the following materials during extraction of sulphur.
- I - Super heated water.  
(1mk)
- II - Hot compressed air.  
(1mk)
- f) (i) Name the process used to manufacture Sulphuric (VI) acid.  
(½mk)

(ii) Calculate the mass of Sulphuric (VI) acid required to react with excess ammonia gas to produce 125.2 tons of ammonium sulphate fertilizer.  
(3mks)

(iii) State the property of Sulphuric (VI) acid that is illustrated by its reaction with sucrose.  
(1mk)

3. The flow chart below shows how nitric (v) acid is produced on a large scale. Study it and answer the questions that follows.



a) State the functions of:

(i) Purifier

(1mk)

(ii) Heat exchanger

(1mk)

b) Identify

(i) Gas A.....

( ½ mk)

(ii) Gas B .....

( ½ mk)

(iii) Catalyst C.....

( ½ mk)

c) Write equations for the reaction that take place.

(i) In catalytic chamber.

(1mk)

(ii) In absorption tower.

(1mk)

d) Calculate the molarity of the commercial nitric (v) acid, given that it is 68% pure and has a density of  $1.42\text{g/cm}^3$ . (N=14, H=1, O=16)

( 2 ½ mks)

e) (i) Complete the table below to show the observations made when concentrated nitric (v) acid is added to the substances shown and warmed.

(2mk)

| Substance                    | Observation |
|------------------------------|-------------|
| Acidified Iron (II) sulphate |             |
| Sulphur powder               |             |

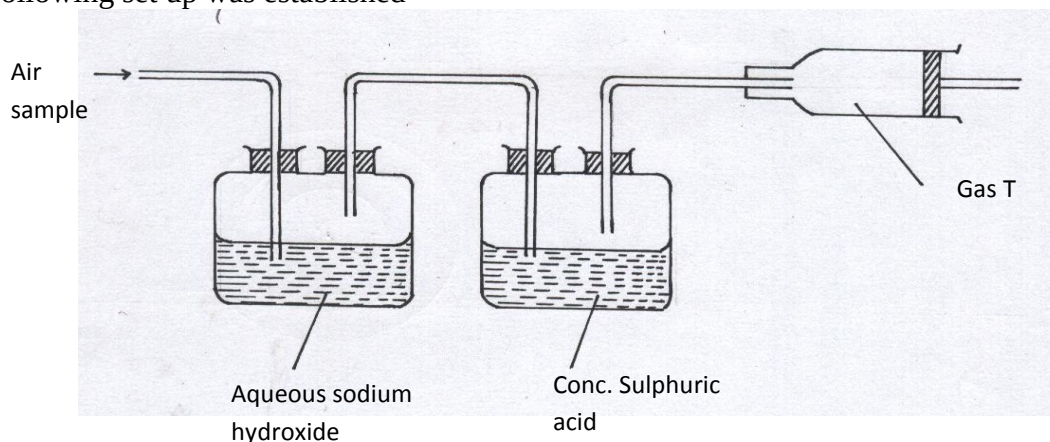
(ii) Give reasons for the observations made using:

I Acidified iron (II) sulphate.

(1mk)

II Sulphur powder  
(1mk)

4. In order to find out the proportion by volume of one of the main constituents of air, the following set up was established



- a) (i) Name two constituents of gas **T**  
(1mk)
- (ii) Suggest a reason for passing air through:
- I Aqueous sodium hydroxide  
(1mk)
- II Concentrated sulphuric acid  
(1mk)
- b) The volume of the gas collected in the syringe was  $60\text{cm}^3$ . This was passed repeatedly over hot copper powder in the combustion tube until no further change of volume took place. When cooled to the original temperature, the volume was reduced to  $47.4\text{cm}^3$
- (i) What observation was made in the combustion tube?  
(1mk)

(ii) Which constituent of air was removed by copper powder?

(1mk)

(iii) Calculate the percentage of the gas in (ii) above in the sample of air

(2 mks)

c) The remaining gas in the syringe was repeatedly passed over hot Magnesium metal in the second combustion tube

(i) Name the main component in  $47.4\text{cm}^3$  of the remaining gas

(1mk)

(ii) Write an equation for the reaction in the second combustion tube

(1mk)

(iii) Name two of the gases still in the syringe at the end of the experiment.

(2 mks)

d) Iron roofing sheets are coated with Zinc as a sacrificial metal

(i) What is meant by the term “sacrificial”

(1mk)

(ii) Give the name given to the process by which iron sheets are coated with Zinc

(1mk)

(iii) Write the equation of the reaction in which Iron rusts

(1mk)

(iii) Zinc is higher than Iron in the reactivity series yet it does not corrode as fast as Iron. Explain (1mk)

5. In an experiment to determine the percentage of impurity in Sodium carbonate, 1.8g of impure Sodium carbonate was reacted with excess 2M Hydrochloric acid. 340cm<sup>3</sup> of dry Carbon (IV) oxide gas was collected during the experiment at room temperature and pressure. (Na=23, O=16, C=12; Molar gas volume at r.t.p=24dm<sup>3</sup>)

a) Why was excess 2M Hydrochloric acid used in the experiment?

(1mk)

b) Write an equation for the reaction that produced Carbon (IV) oxide

(1mk)

c) Calculate

i. The number of moles of Carbon (IV) oxide produced

(2mks)

ii. The number of moles of Sodium carbonate that reacted with the acid

(2mks)

iii. The mass of Sodium carbonate that reacted with the acid

(2mks)

iv. The percentage of impurities in the sample of Sodium carbonate

(2mks)

6. a) Write down the structural formula of the following compounds

(i) 2, 2 – Dimethylpropane

( 1 mk)

(ii) 2 – Chloropropene

( 1 mk)

iii) Tetrachloro methane

(1 mk)

b) A,B,C are three homologous series of organic compounds

| Series | General formula |
|--------|-----------------|
| A      | $C_nH_{2n-2}$   |
| B      | $C_nH_{2n}$     |
| C      | $C_nH_{2n+2}$   |

(i) What is the name given to series C

(1 mk)

(ii) Write down the name and structural formula of the third member of series “B”

(2mks)

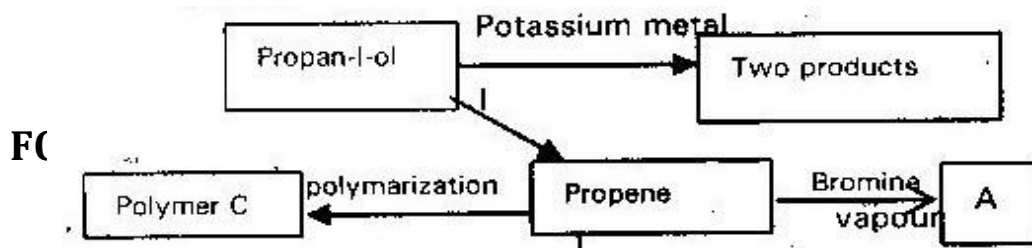
Name: .....

Structure:

(iii) Write down an equation and name the products of reaction between HBr with second member of series “B”

(2 mks)

c) Study the scheme given and answer the questions that follow



- i. Write an equation for the reaction in process II ( 1mk)
- ii. Name process I and II ( 2 mks)
- iii. I..... II.....  
Identify the products “A” and “B” ( 2 mks)
- iv. A..... B.....  
Name ONE catalyst used in process II (1 mk)



# FORM 3 MIDTERM 3 EXAMS

## BUSINESS STUDIES

### PAPER 1

NAME .....ADM.....CLASS.....

**Answer all the question in the spaces provided after each**

1. Outline four circumstances under which a business person would allow a customer to acquire goods on credit. (4mks)

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2. State any four errors that will not affect the balancing of a trial balance. (4mks)

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3. Highlight four demerits of using machines in carrying out office activities. (4mks)

**FOR MARKING SCHEMES CALL/TEXT/WHATSAPP 0705525657**

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4. Name the factor that each of the following resources relate to (4mks)

| Resource   | Factor of production |
|------------|----------------------|
| a. Manager |                      |
| b. Water   |                      |
| c. Owner   |                      |
| d. Vehicle |                      |

5. Outline four ways in which warehousing promotes trade in Kenya.  
(4 marks)

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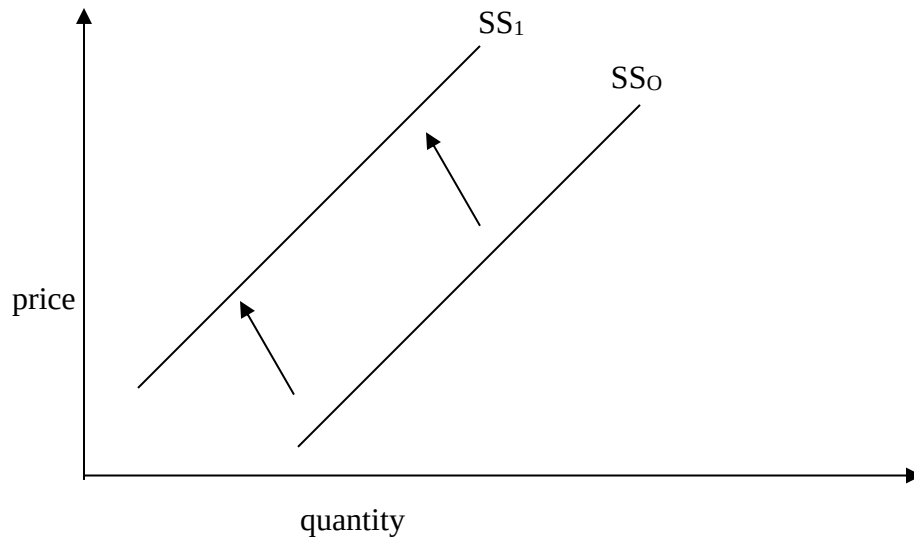
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6. The diagram below shows a shift in supply curve from  $S_0S_0$  to  $S_1S_1$



Identify four factors that have made the supply curve to shift from  $S_0S_0$  to  $S_1S_1$  (4mks)

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7. State the type of advertising that each of the following statements relate to (4mks)

|      | Statement                                | Type of advertising |
|------|------------------------------------------|---------------------|
| i.   | Persuade consumers to buy a product      |                     |
| ii.  | Promotes a particular brand of a product |                     |
| iii. | Create awareness about a product         |                     |
| iv.  | Promotes the name of the manufacturer    |                     |

8. For each of the following transactions, state the account to be debited and the account to be credited. (4mks)

| Transaction | Account to debit | Account to credit |
|-------------|------------------|-------------------|
|-------------|------------------|-------------------|

|    |                                     |  |  |
|----|-------------------------------------|--|--|
| a. | Paid a creditor from private source |  |  |
| b. | Brought in cash from private source |  |  |
| c. | Withdrew cash for personal use      |  |  |
| d. | Bought goods and paid by cheque     |  |  |

9. State four reasons why consumers have to make a choice between competing needs.  
(4mks)

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10. State four limitations of containerization. (4mks)

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11. Highlight four ways in which prices of goods and services may be determined in the market other than the forces of demand and supply. (4mks)

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12. Outline four uses of balance sheet by a business.

(4mks)

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13. Outline four reasons why government may find it necessary to protect consumers. (4mks)

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14. State four features that distinguish a private company from a public company.(4mks)

| Private company | Public company |
|-----------------|----------------|
| .....<br>.....  | .....<br>..... |
| .....<br>.....  | .....<br>..... |

|    |  |
|----|--|
| 6. |  |
| 7. |  |

15. Highlight four benefits that may accrue to a customer who buy goods from a manufacturer (4mks)

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16. State four circumstances under which a seller would issue a credit note to a buyer (4mks)

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17. Outline four roles of an entrepreneur in a business. (4mks)

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18. Illustrate with simple diagram, four channels a local farmer would use to distribute his produce.(4 mks)

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19. Outline four benefits that will accrue to a firm as a result of increased scale of operation(4mks)

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20. Give two examples of products that would face the following types of demand.(4mks)

a) Derived demand

b) Joint demand

- c) Composite demand
- d) Competitive demand

21. Outline four roles of filing in an office. (4mks)

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22. Outline four effects of an ageing population on an economy. (4mks)

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23. Indicate the balance sheet items and show how they would be affected by the following transactions. (4mks)

- a. Purchase of a motor bicycle by cheque
- b. Sale of goods on credit to Baraka
- c. A debtor pays his debt in cash



- d. The owner converted his personal house into business use

**24.** The following trial balance was incorrectly prepared  
**KALWAL TRADERS**  
**TRIAL BALANCE**  
**AS at 31-Dec-2004**

|                 | DR           | CR           |
|-----------------|--------------|--------------|
| Capital         |              | 31000        |
| Cash in hand    | 11000        |              |
| Creditors       | 14000        |              |
| Insurance       |              | 12000        |
| Debtors         | 13000        |              |
| Drawings        |              | 15000        |
| Bank over draft |              | 6000         |
| Bank loan       | 18000        |              |
| Stock           |              | 18000        |
|                 | <u>56000</u> | <u>56000</u> |

Required: Prepare the correct trial balance  
 (4 marks)

25. Outline four differences between a monopoly and a monopolistic competition as product market (4mks)

| Monopoly | Monopolistic competition |
|----------|--------------------------|
| i.       |                          |
| ii.      |                          |
| iii.     |                          |
| iv.      |                          |

# FORM 3 MIDTERM 3 EXAMS

## BUSINESS STUDIES

### PAPER 2

NAME

.....ADM.....CLASS.....

*Answer any five questions*

1. (a) Explain five principles of insurance (10mks)  
(b) Explain five ways through which containers may be in effective as a way of moving goods (10mks)
2. (a) Explain five internal economies of scale that may be experienced by the firm (10mks)  
(b) Explain five factors to be considered when setting up an office layout for a business organization (10mks)
3. (a) Explain five causes of unemployment (10mks)  
(b) Explain five benefits of preparing a business plan to a firm (10 marks)
4. (a) Explain five factors that influence choice of product promotion (10mks)  
(b) Explain five roles played by commercial attaches in trade promotion (10 marks)
5. (a) Describe five differences between a movement along a demand curve and a shift of a demand curve (10mks)  
(b) The following information relates to Maralal Traders for the month of May 2014.  
May 1: Balance brought forward:  
Cash Shs. 180,000  
Bank Shs. 450,000 (Cr)  
3: Received a cheque of 1,500,000 from Mpasha, a debtor  
7: Cash sales Shs. 280,000  
11: Jolloimat, a creditor of Shs.600, 000 was paid by cheque of sh 400,000 and the balance by cash.  
14: Received commission in cash sh. 150,000  
17: A debtor, Leteipa, paid his account of 185,000 by cheque less 2% cash discount.  
19: Paid the following expenses by cheque:  
Rent Shs. 75,000  
Electricity Shs. 32,000  
Water Shs. 25,000  
21: Withdrew Shs. 100,000 from bank for personal use  
24: Paid cash Shs 133,280 to Kinai after deducting a cash discount of 2%  
25: Received a cheque of sh. 200,000 from Kiyapi, a debtor  
28: Paid Salaries Shs. 120,000 by cheque.  
31: Banked all cash except sh. 50,000.

**Required:**

Record the above transactions in a three column cash book.

(10marks)

**FOR MARKING SCHEMES CALL/TEXT/WHATSAPP 0705525657**

6. (a) Explain five reasons that may limit use of pipeline transport in a country **(10mks)**

(b) Kavetsa had the following balances on 1<sup>st</sup> January 2010

|               |         |  |
|---------------|---------|--|
|               | sh      |  |
| Motor vehicle | 210,000 |  |
| Debtors       | 20,000  |  |
| Cash          | 90,000  |  |
| Stock         | 25,000  |  |
| Creditors     | 40,000  |  |
| Cash at Bank  | 50,000  |  |

During the month, the following transactions took place

2010 January 4: paid sh 12,000 to creditors by cheque

“ 13: purchased stock worth sh 18,000 on credit

“ 26 : deposited sh 6,000 into the business bank account from cash till

Record the above transactions in the relevant ledger accounts and balance them off **(10mks )**

# FORM 3 MIDTERM 3 EXAMS

NAME: .....ADM. NO: .....CLASS:...

## AGRICULTURE PAPER 1

### INSTRUCTIONS:

Answer ALL Questions in section A, B and any TWO Questions in section C.

### SECTION A: (30 MARKS)

1. State four characteristics of large scale farming system. (2mks)
  
  
  
  
  
  
  
  
  
  
2. Outline four factors that can make shifting cultivation practicable. (2mks)
  
  
  
  
  
  
  
  
  
  
3. State two effects of high temperature on crop production. (1mk)
  
  
  
  
  
  
  
  
  
  
4. State two effects of sub-soiling in land preparation. (1mk)
  
  
  
  
  
  
  
  
  
  
5. Differentiate between soil texture and soil structure. (2mks)
  
  
  
  
  
  
  
  
  
  
6. State two reasons why green manure is not commonly used by many farmers. (1mk)

**FOR MARKING SCHEMES CALL/TEXT/WHATSAPP 0705525657**

7. Enumerate four types of farm records kept by farmers. (2mks)

8. Outline two importance of a title deed in land tenure system. (1mk)

9. State reasons why it is difficult to control the following weeds. (4mks)

i) Oxalis –

ii) Couch grass –

iii) Black jack –

iv) Nut grass –

10. What is the meaning of the following terms as used in pest control?

i) Economic injury level –  
(1mk)

ii) Integrated pest management –  
(1mk)

11. State two varieties of beans growth in Kenya. (2mks)

12. List two advantages of a grass-legume pasture over pure stand grass pasture.  
(1mk)

13. Why is too much air undesirable in silage making? (1mk)
14. State four reasons for staking of tomatoes. (2mk)
15. Outline two methods of breaking seed dormancy. (1mk)
16. List two demerits of using seeds as planting materials. (1mk)
17. Give two importance of raising seedlings in polythene sleeves compared to direct establishment on the ground. (1mk)
18. (a) State three post harvesting practices carried out in crop production. (3mks)
- (b) List two limitations of using a traditional granary in crop storage. (2mks)

**SECTION B: (20MARKS)**

**Answer ALL Questions in this section.**

19. Below are diagrams of common weeds found in the farm. Use them to answer questions that follow.

Below are diagrams of common weeds found in the farm. Study them carefully.

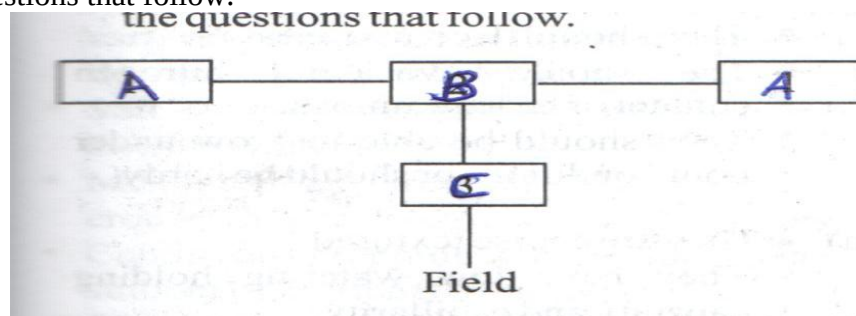


i) Identify weeds Q, R and S. (3mks)

ii) Mention one harmful effect of weed Q and R. (1mk)

iii) Give a reason why weed S is referred to as a parasitic weed. (1mk)

20. The diagram below is an illustration of turning a certain type of manure. Use it to answer questions that follow.



a) Name the type of manure that is turned using the above method. (1mk)

b) By use of arrows, show how the manure is turned. (1mk)



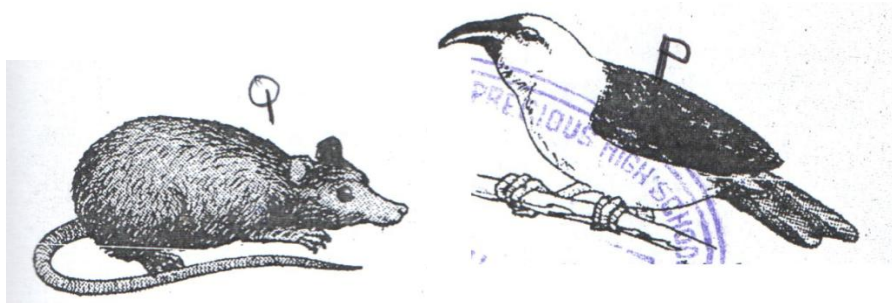
c) What is the significance of adding the following during the preparation of compost manure?

i) Adding well rotten manure. (1mk)

ii) Adding garden soil. (1mk)

iii) Adding ash. (1mk)

21. Below are representations of certain pests that attack crops. Use them to answer questions that follow.



i) Identify pests P and Q. (2mks)

ii) State two effects of pest Q in crop production. (2mks)

iii) Highlight three methods of controlling pest P. (3mks)

iv) At what stage do pest P attack crops? (1mk)

**SECTION C: (40 MARKS)**

**Answer any TWO Questions.**

22. (a) Explain five effects of soil erosion. (10mks)  
(b) Explain five methods used to control crop diseases. (10mks)
23. Describe production of maize under the following:
- a) Ecological requirements. (3mks)
  - b) Seed bed preparation. (4mks)
  - c) Planting. (5mks)
  - d) Field management practices. (5mks)
  - e) Harvesting. (3mks)
24. (a) Describe ten (10) nursery management practices that are carried out after seed germination. (10mks)  
(b) Explain five factors to be considered when designing a crop rotational programme. (10mks)

# FORM 3 MIDTERM 3 EXAMS

NAME: .....ADM. NO: .....CLASS:...

## AGRICULTURE PAPER 2

### INSTRUCTIONS:

Answer ALL Questions in section A, B and any TWO Questions in section C.

### SECTION A: (30 MARKS)

1. List two biotic factors that affect livestock production in Kenya. (1mk)
  
2. Outline four reasons that enable camels to live in arid areas. (2mks)
  
3. State four reasons for dehorning in livestock. (2mks)
  
4. State four advantages of using a spray race over a plunge dip. (2mks)
  
5. State four reasons for seasoning timber. (2mks)

- 
6. State two types of bees. (2mks)
7. State four reasons for swarming of bees. (2mks)
8. Define the term caponization as used in livestock production. (1mk)
9. Outline four uses of water in animals on the farm. (2mks)
10. Differentiate between steaming up and creep feeding. (2mks)
11. State four routes through which pathogens enter the body of an animal. (2mks)
12. Differentiate between carrying capacity and overstocking. (2mks)
13. State four advantages of Zero grazing unit in livestock production. (2mks)
14. Name the equipment used alongside each of the following: (2mks)
- i) Trocar –
  - ii) Bull ring –
  - iii) Elastrator –
  - iv) Syringe –

15. State four reasons that show that a sow is about to farrow. (2mks)

16. State two major systems of breeding in livestock production. (2mks)

**SECTION B: (20 MARKS)**

**Answer ALL questions in this section.**

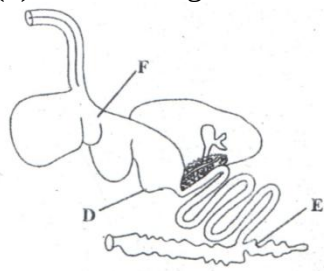
17. Mr. Gatimu asked his daughter to mix feeds for his cows. Mr. Gatimu's daughter decided to use feedstuffs to get 15% DCP. The contents were as shown below

- Fish meal 50% DCP.
- Cereal meal mixture 12% DCP

(a) How much of the two feedstuffs will Mr. Gatimu's daughter mix to get a 90kg bag?  
(4mks)

(b) The diagram below illustrates a type of a digestive system.

i) Name parts marked D and E (2mks)



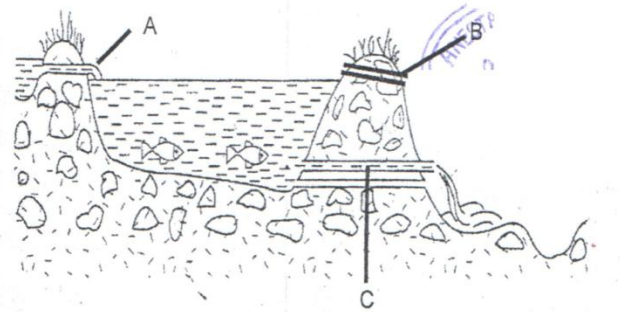
D.....

E.....

ii) State two functions of part F.

(2mks)

18. The diagram below shows a cross-section through a fish pond. Study it and answer questions that follow.



a) Name parts A, B and C. (1 ½ mks)

b) What is the function of part labeled B. (½ mk)

c) State four factors that should be considered when siting a fish pond.(2mks)

19. State four categories in which diseases in livestock are classified. (2mks)

20. Below is an illustration of a livestock parasite. Study it and answer questions that follow.

i) i)Name the parasite( ½ mk)



ii) State three symptoms that may be observed in an animal that has been attacked by the above parasite. (1 ½ mks)

iii) State the intermediate host of the parasite. (½ mk)

iv) Give two control measures of the above parasite. (2mks)

21. State two methods of extracting honey from combs. (2mks)

**SECTION C: (40 MARKS)**

**Answer any TWO questions from this section.**

22. (a) Describe five factors to consider when selecting a breeding stock. (10mks)

(b) Describe five control measures of diseases in livestock and give an example of a disease controlled by the measure. (10mks)

23. Discuss mastitis disease under the following:-

- a) Causal organism. (1mk)
- b) Symptoms. (5mks)
- c) Causes. (4mks)
- d) Control measures. (5mks)
- e) State five reasons for keeping livestock healthy. (5mks)

24. (a) Explain five reasons why fences are important to a farmer. (10mks)

(b) Explain five modern technology methods that have helped to improve the quality of livestock products in Kenya. (10mks)

# FORM 3 MIDTERM 3 EXAMS

NAME: .....ADM. NO:  
.....CLASS:.....

## CHRISTIAN RELIGIOUS EDUCATION PAPER ONE

1. a) Name **six** books that belong to minor prophets.  
(6mks)  
b) Give **eight** reasons why the Bible was translated into different languages .  
(8mks)  
c) Outline **six** ways in which the teaching of Christian Religious Education enhances national unity.  
(6mks)
2. a) Outline **eight** ways in which Abraham demonstrated his faith in God.  
(8mks)  
b) State **six** instructions of God to Abraham regarding circumcision.  
(6mks)  
c) Explain **six** factors that have undermined traditional African circumcision  
(6mks)
3. a) State **seven** characteristics of the local Canaanite religion.  
(7mks)  
b) Explain **seven** forms of punishment prophesized by Elijah on King Ahab and Jezebel.  
(7mks)  
c) Give **six** reasons why killing was condemned in traditional African communities.  
(6mks)
4. Explain **five** reasons why God sent prophets in Israel  
(10 mks)  
b) State **six** teachings of prophet Amos on hypocritical religion in Israel.  
(6mks)  
c) State **four** lessons Christians leant about God from the visions of Prophet Amos.  
(4mks)



5. a) Give **six** reasons why the rebuilding of the wall of Jerusalem was important to Nehemiah's people.  
(6mks)
- b) Explain the suffering and lamentations of Jeremiah  
(7mks)
- c) Give **seven** reasons why it is difficult to have reforms in Kenya.  
(7mks)
6. a) Explain factors which promote harmony and mutual responsibility in Traditional African Communities.  
(8marks)
- b) Identify **six** importance of observing taboos in African traditional Communities? (6 marks)
- c) Identify **six** factors which have led to decline in observance of taboos in Traditional African Communities. (6 marks)

# FORM 3 MIDTERM 3 EXAMS

NAME .....ADM.....

## CREPAPER 2

**ANSWER ANY FIVE QUESTION S IN THE ANSWER SHEET PROVIDED**

1. With reference to Luke's gospels, explain eight ways in which Jesus fulfilled the Old Testament prophecies about the Messiah. (8mks)
  - b) Explain What Zachariah revealed about John the Baptist in the Benedict sin Luke 1:68-80 (6mks)
  - c) Give reasons why then birth of a child is celebrated in the modern society? (6mks)
- 2 a) Give reasons why Jesus was rejected in Nazareth (8mks)
  - b) Describe the healing of the paralytic (lk 5:17-26) (7mks)
  - c) Give reasons why a church leader may be rejected today (5mks)
- 3 a) Narrate the parable of the widow and the unjust judge ( lk18:1-8) (6mks)
  - b) Explain Jesus teaching on prayer (8mks)
  - c) Give the importance of prayers in a Christian life today (6mks)
- 4a) Explain actions taken by Jewish leaders to ensure that Jesus was put to death (8mks)
  - b) Give evidence to show that Jesus resurrected (6mks)
  - c) What is the importance of Jesus resurrection to Christian today? (6mks)\_
- 5a) Explain the New Testament teaching s on the unity of the believers as expressed in the bride (8mks)
  - b) Identify causes of disunity in the early church (6mks)
  - c) Explain how church disciplines those who cause disunity in to church today (6mks)
- 6 a) Outline seven teachings about Jesus from peters message on the day of Pentecost (7mks)
  - b) Outline the characteristics of love as taught by St.Paul in 1corinthian 13 (7mks)
  - c) State activities of the church in Kenya which show that the Holy Spirit is working among them. (6 marks)

**FOR MARKING SCHEMES CALL/TEXT/WHATSAPP 0705525657**

# FORM 3 MIDTERM 3 EXAMS

NAME .....ADM NO ..... CLASS  
.....

**101/1 ENGLISH PAPER 1 (FUNCTIONAL SKILLS, CLOZE TEST AND ORAL SKILLS ) TIME 2HRS**

## **1.FUNCTIONAL WRITING**

Imagine that you are organizing secretary of Ruth Kerubo's Graduation Party. Ruth has just graduated with a First Class Honour's from Kenyatta University.

- (a)Design an invitation card to be sent to the guests .(10mks)
- b) Write a congratulatory note to Ruth to be presented that day (10mks)

## **2.CLOZE TEST (10mks)**

**Read the passage below and fill in each blank space with the most appropriate word .**

Passing National Examination in Kenya has become a matter 1\_\_\_\_\_. This is primarily because , more often than not , examination 2\_\_\_\_\_ the future of the student . From our collective experience this is very clear . A standard eight pupil must 3\_\_\_\_\_ well secure a place in a good 4\_\_\_\_\_ school . Similarly , a form four candidate must not just pass , but must also be among the students who score public 5 \_\_\_\_\_. The reverse often has painful consequences for the student , 6\_\_\_\_\_ means falling by the wayside and in a country with limited employment 7 \_\_\_\_\_ , chances of a decent livelihood become severely compromised .It is precisely because of those stark realities that students find 8\_\_\_\_\_ under seething pressure to pass 9\_\_\_\_\_. On the other hand , parents who know all too well that failing to score good grades could mean that the end of the road of their children's 10\_\_\_\_\_ are willing to cut all corners to ensure the children pass their exams.

## **3.ORAL SKILLS (30mks)**

- (a) Read the narrative below and answer the questions that follow.

### **THE COCK AND THE KITE**

(The setting of the story is in Kibiro , Uganda in the Western Rift Valley near Lake Albert )

A long time ago , there lived cock and his family as well as kite and his family .The former was hard working while the latter was lazy .It happened that the place was hit by a famine .People from far used to travel a long way to go to Kibiro to barter food for salt .It also happened that both families ran out of salt . Cock's wife informed her husband that they had ran out of salt

**FOR MARKING SCHEMES CALL/TEXT/WHATSAPP 0705525657**

and asked him to take some finger millet to Kibiro .He agreed , went to Kibiro , obtained salt and set upon the return journey .

The other family got wind of this .Mrs.Kite also asked her husband her husband to go to Kibiro and try to get salt since the lazy family did not have anything to take to Kibiro .On the way , he met the cock resting on his way home with the salt behind him .He was standing on one leg having hidden one of his leg in his wing , as cocks do many times when resting .Kite asked cock how he managed to get the salt , where upon cock told kite that the salt miners ad cut off one of his legs in exchange for the salt .Kite accepted the lie and proceeded towards Kibiro ready to do the same .Cock continued on his journey and got home safely .

On arrival at Kibiro, Kite offered his leg for a bundle of salt which the miners readily accepted .His leg was consequently amputated , rendering him immobile , even unable to carry home salt poor kite flew back home, where he was received by his family in much grief , especially when he narrated to them the ordeal he went through . Later , kite's family was to receive the traumatizing news that cock had actually ill-advanced Kite , leading to loss of his leg .

Hence forward , great enmity ensued between the two families with Kite's family swearing to retaliate by hunting Cock's family down and eat them. This goes -+on to date.

(i)The narrator notices signs of in attentiveness among the audience during the story telling session What could be the problem ? (4mks)

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(ii) Explain how you would make the narration of the first two paragraphs effective .(3mks)

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(iii)Mention three ways in which you would know that your audience in this story is fully participating in the performance . (3mks)

.....

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.....  
.....  
(b) Underline the silent letter in the words below

(4mks)

Pseudo

Subtle

Deign

Damn

(c) Explain the meaning brought out by stressing the underlined word in each of the following sentences . (3mks)

(i) Muriithi spoke to Gatwiri yesterday .

.....  
.....  
.....  
(ii) Muriithi spoke to Gatwiri yesterday .

.....  
.....  
.....  
(iii) Muriithi spoke to Gatwiri yesterday.

.....  
(d) Read the following conversation and answer the questions that follow. (7mks)

MBAIRE: hi Mr. Katana , long time no see.

MR .KATANA: hello Mbaire , how have you been for so long ?

MBAIRE: I'm fit as you can see .

MR.KATANA : What a surprise to see you here! Do you live around this area?

MBAIRE: Hi , just popped in to have a glimpse of some associates of mine . And you ?

MR.KATANA : Well , I came to visit a colleague who has been ailing for some time . You remember Mr. Kwach ?

MBAIRE: Yes , the leopard ! Who can forget him ? He used to .....

MR.KATANA: Well , I must be going . Goodbye.

MBAIRE:See you.

(i)Identify one short coming in Mbaire’s responses.(2mks)

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(ii)Give four aspects of speech that Mbaire needs to consider so as to communicate effectively and in an appropriate manner.(4mks)

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(iii)Give two possible reasons for Mr. .Katana’s exit before Mbaire finishes speaking .(2mks)

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.....

(e)Write another word pronounced the same way as the word given below. (3mks)

Call-

Awe-

(f)In the words given below .Underline the part that should be stressed.(2mks)

(i)Excuse (verb)

(ii)Reject (noun)

# FORM 3 MIDTERM 3 EXAMS

## ENGLISH

### PAPER 2

**1. Read the passage below and then answer the questions that follow.**

#### **THE DETOX DEBATE**

Detox kits and supplements are recent health fads. But can they really help you lose weight fast? Online weight loss coach Andrew Cate investigates.

We've heard a lot about the need to detox lately. It is the term used to describe a strict program of elimination and supplementation that is meant to rid your body of impurities, cleaning your liver and kidneys, and flushing your bowel. It is suggested that toxins build up from consuming too much fat, sugar, alcohol, caffeine, preservatives and pollution.

There's no shortage of detox books, kit and programs claiming to help you shed weight, improve your well-being, cause your skin to radiate and make you feel younger. The kits usually contains a dietary program which is supplemented with a variety of vitamins, minerals, tonics, digestive aids and laxatives. They are particularly popular in January as people feel the urge to begin the New Year afresh after overindulging during the festive season.

People will make drastic changes when they go on a detox diet and often feel better for starting a structured regime. However, detox kits made up of herbal laxatives and diuretics are unnecessary and have generally no proven benefit. Detox regimes do not improve kidneys and liver function. These organs are designed to filter your blood fats, alcohol and other nasties- all without the help of a fancy box from your local pharmacy. There is no scientific evidence to support specific detox diets. Programs or supplement kits. However, there's no debate about the fact that eating less junk food, cutting out cigarettes and your alcohol intake etc will benefit your health. For example drinking more water and cutting out caffeine will improve your bowel function. These changes will enhance your well-being but there's nothing magical about the detox diet itself. Rather it's the associated lifestyle changes that benefit your health.

Detox kits that contains laxatives and diuretics to encourage you to fast could, potentially, do more harm than good. Laxatives speed up your bowel motions, but also prevent absorption of nutrients, while diuretics can result to partial dehydration.

The fasting component of a detox should only be minimal, and not extend beyond a day or two. By eating next to nothing, you are not getting enough nutrients for the essential functions of your body. Supplements are no substitute for real food, and relying solely on them can result in vitamin deficiencies. Fasting is also known to slow down your metabolic rate, which encourages your body to store fat, making it harder to lose body fat in the future.

If you've spent weeks, months or years overindulging, drinking and smoking you can't hope to fix yourself in a few days. Detox diets aren't an instant cure to health and wellness. Short-term changes to your diet and lifestyle only result in short-term changes to your health and wellness. If you can't stick to the lifestyle changes you make over the long term, there's no point starting them as they won't have any serious impact upon your health.

**Questions.**

- a) What is detoxing? (2marks)
- b) From the information given in the passage, what builds up toxins in the body? (2mks)
- c) Give the contents of a detox kit. (2 marks)
- d) When do detox kits sell most? (1 mark)
- e) In about 80 words summarize the writer's argument on whether we need to detox or not. (5marks)



- f) Outline the dangers of detox diets. (3marks)
- g) Detox diets aren't an instant cure to health and wellness. (Write in the affirmative)(1 mark)
- h) We've heard a lot about the need to detox lately. (Re-write the sentence into passive)(1 mark)
- i) Explain the meaning of the following words as used in the passage.(3marks)
- I. Laxatives
  - II. Overindulging
  - III. Deficiencies

**2. Read the following excerpt and answer the questions that follow.**

Her dream was rudely and violently interrupted by a thunderous bang and a loud roar of laughter. She woke up with a start, jumped up to her feet and stared at the door with wide panic-stricken eyes. For a moment she could not figure out her surroundings and called out the name of her sister Taiyo. She was terrified. The door flung open and Olarinkoi staggered in. He was stone drunk. Resian stared at him **unblinkingly** as he walked towards her and she backed off terrified, squeezing herself flatly against the wall. He followed her there and got hold of her shoulders and shook her violently glaring at her with his glittering eyes.

"You silly thing," he thundered angrily. "I tell you to prepare food and you refuse to do so, eh? Today you will know who is the owner of this home. If you are still in doubt, let me tell you frankly that from today on you are my wife, hear that, eh? You are my wife. For a long time you

have been sneering at me, showing how highly educated you are. Today we shall see how educated your body is! Yes, we shall see!’’

He got hold of her hand and began dragging her into the other room. At first she did not understand his intention until he began unfastening her buttons with his rough trembling hands. Then the truth came, and with it, terror and panic. She tried to get away from him, but he held her **effortlessly** as he brutally continued fumbling with her dress, trying to loosen it. She screamed as loudly as she could while she pushed him away and thrashed frantically about. But that did not deter him and he totally ignored her screams holding her more firmly with his strong arms. Against her loud protest, he tore her garments and began to push her towards the bed. Then desperately she took the last chance of self-defense and self-preservation. **Mustering** all her strength, she thrust his thumb into her mouth; sunk her teeth into the flesh like a ferocious animal and tenaciously held onto it, tugging at it fiercely like a lioness. She could feel the flesh tearing and she tasted the salt of his blood as it filled her mouth but she clung unto the thumb as Olarinkoi **howled** with pain.

- a) What are the contents of Resian’s dream? (4 marks)
  
- b) Discuss one character trait of Resian in this excerpt.(2 marks)
  
- c) What has greatly changed in Olarinkoi in this excerpt compared to his earlier conduct in the novel? (3 marks)
  
- d) Identify and illustrate two aspects of style in this excerpt.(4 marks)
  
- e) Explain two themes in the excerpt. (4 marks)

f) Explain what happens immediately after the excerpt.(3 marks)

g) “Today we shall see how educated your body is.” Rewrite in reported speech. (1 mark)

h) Explain the meaning of the following expressions in the passage. (4 marks)

(i) Unblinkingly

(ii) Effortlessly

(iii) Mustering

(iv)Howled

**3. Read the following oral poem and then answer the questions that follow.**

Listen

My husband

In the wisdom of Lang’o

Time is not stupidity split up

Into seconds and minutes.

It does not follow

Like beer in a pot

That is sucked

Until it is finished.

It does not resemble

A loaf of millet bread

Surrounded by hungry youths

From a hunt

It does not get finished  
Like vegetable in the dish.

A lazy youth is rebuked  
A lazy girl is slapped  
A lazy wife beaten  
A lazy man laughed at  
Not because they waste time  
But because they only destroy  
And do not produce

And famine  
Invades your villages  
And women take their baskets  
To go and beg, food,  
Strangers will sleep with them.  
They will have your wives  
And what can you say?

### Questions

- a) Classify the above song. (2 marks)
- b) Who is the singer? (2 marks)
- c) Explain two economic practices by the community from which the song is derived. (4 marks)
- d) Identify and illustrate three aspects of style found in this song. (6 marks)

- e) Contrast the Africans and Westerners in view of: (2 marks)
- Time

- Concept of hard work

- f) Giving examples, state the dominant tone of this song. (2 marks)

- g) Identify the moral lesson in this song. (2 marks)

### GRAMMAR

**a) Answer the following questions according to the instructions given after each. (3marks)**

- I. In the 16<sup>th</sup> century, shake spears wrote many plays. (Underline the adverbial phrase)
- II. If you come to the arena, you will meet the great Isukuti dancers. (Re-write using.... Unless)
- III. The principal advised the students to consider the consequences of their behavior. (Rewrite in direct speech)

**b) Complete each of the following with the correct phrasal verb formed from the word given in brackets. (2 marks)**

- I. Don't worry, the police will \_\_\_\_\_ the matter. (look)  
II. Njoroge is not reliable. He promised to support our plan but \_\_\_\_\_ at the last minute. (back)

**c) Use the correct form of the words in brackets to fill the blanks. (2 marks)**

- I. The inmates took advantage of the \_\_\_\_\_ to escape from custody. (confuse)  
II. The matter should be handled \_\_\_\_\_. (caution)

**d) Fill in the blank with the appropriate choice between the two given in brackets. (3 marks)**

- I. No criterion \_\_\_\_\_ been established. (has/have)  
II. The mother had just \_\_\_\_\_ down to rest when the baby cried. (laid/lain)  
III. The number of excellent English books in our library \_\_\_\_\_ increased. (has/ have)

**e) Join the following pairs of sentences by using a relative pronoun. (2 marks)**

- I. Nyaga walked along the path. The path led to the river.  
II. Rabies is a disease. It is commonly spread by dogs.

**f) Explain the meaning of the underlined idiom. (1 mark)**

The chairperson ate the humble pie and returned to the negotiation table.

**g) Explain the difference between the following sentences. (2 marks)**

- i. I like you better than him.

III. I like you better than he.

# FORM 3 MIDTERM 3 EXAMS

NAME:.....CLASS:.....ADM NO:.....

## ENGLISH

PAPER 3

TIME 2½HRS

1. **COMPULSORY: IMAGINATIVE COMPOSITION (20 MARKS)**

Either,

- a) Write a story beginning with the words:

I had not thought it was a big problem until I got involved .....

Or

- b) Discuss the measures you would take to curb flooding in our Kenyan urban areas.

2. **The Compulsory Set Text.**

“Self-interest is a vice that whoever engages in it is bound to fail.” Using *Blossoms of the Savannah*, write an essay to support this assertion. (20 marks)

**3. THE COMPULSORY SET TEXT**

**A DOLL’S HOUSE** by **Henrik Ibsen**

(20 marks)

“Love all, trust none.” Show how this is portrayed using illustrations from the play, “A

# FORM 3 MIDTERM 3 EXAMS

NAME:.....CLASS:.....ADM

NO:.....

312/1

## GEOGRAPHY

### PAPER 1

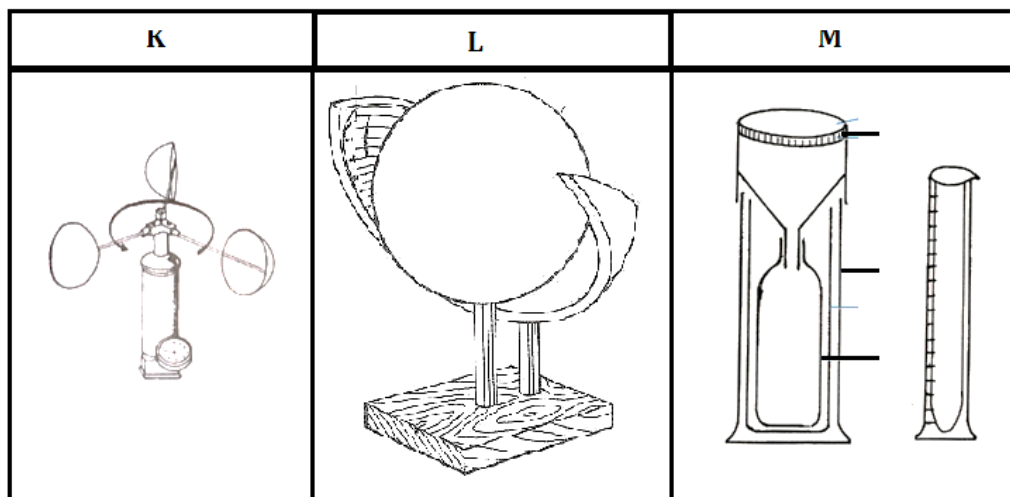
TIME: 2<sup>3</sup>/<sub>4</sub> HOURS

#### INSTRUCTIONS

- This paper two sections: A and B
- Answer all the questions in section A.
- In section B answer question 6 and any other two questions.
- All answers must be written in the foolscaps provided.
- Candidates must answer the questions in English.

#### SECTION A

1. The diagram below shows some weather station instruments.



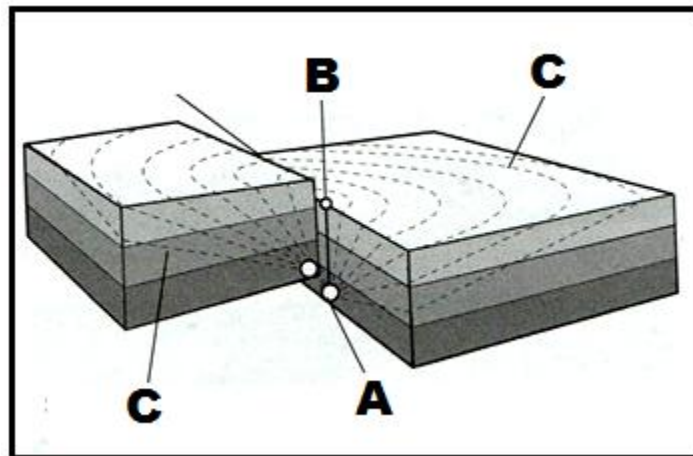
- (a) Identify the instruments marked K, L and M.  
(marks)

(3)



- (b) Outline **two** modern methods of weather forecasting. (2 marks)
2. (a) Give the **two** main movements of the earth. (2 marks)
- (b) State **three** reasons why the interior of the earth is very hot. (3 marks)

3. The diagram below show the occurrence of an earthquake



- (a) Name the parts marked A, B and C (3marks)
- (b) List **two** scales used to measure the magnitude of an earthquake. (2 marks)
4. State **five** characteristics of coniferous forests. (5 marks)
5. (a) What is soil catena? (2 marks)

- (b) State **three** factors that influence soil colour. (3 marks)

### SECTION B

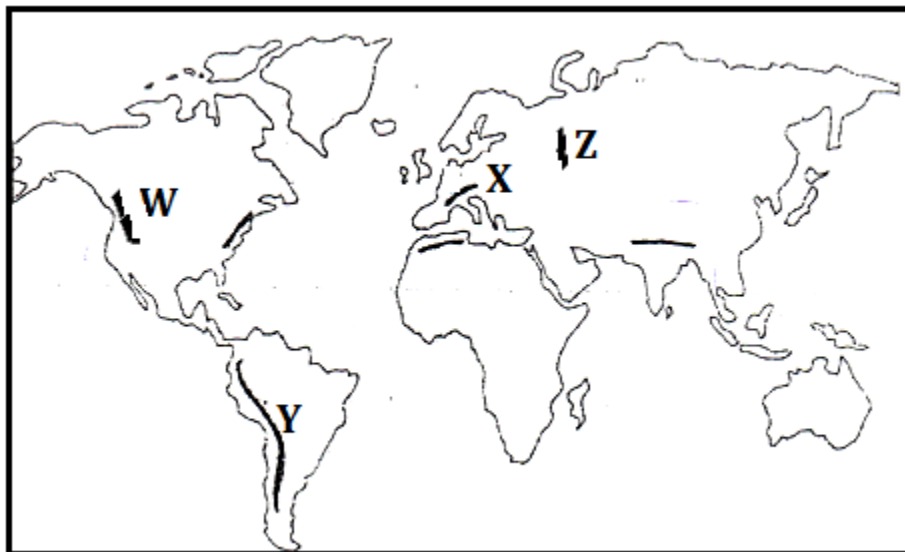
6. Study the map of Oyugis (1:50, 000, sheet 130/1) provided and use it to answer the following questions.

- (a) (i) Identify **two** relief features at grid square 7342. (2 marks)
- (ii) Measure the distance of road E 212 from its junction with road C 18 to the junction at grid square 7542. Give your answer in kilometres. (2 marks)
- (iii) Determine the six figure grid reference of Kitweru School. (2 marks)
- (b) (i) What is the vertical interval of the map? (1 mark)
- (ii) Determine the bearing of a jaggery factory at grid reference 788237 from a church at grid reference 806188. (2 marks)
- (c) (i) Draw a square 10cm by 10cm to represent the region east of easting 85 from northing 30 to northing 40. On the square, mark and name.
- Koderu forest.
  - Oriang' hill
  - A district boundary.
  - All weather road-bound surface.
  - A seasonal swamp
- (6 marks)
- (ii) What is the scale of the sketch that you have drawn? (1 mark)
- (d) (i) Citing evidence from the map, identify **four** crops grown in the area covered by the map. (4 marks)
- (ii) Describe the distribution of settlement in the area covered by the map. (5 marks)
7. (a) (i) What is a mineral? (2 marks)

- marks) (ii) Give **three** ways used to classify rocks. (3 marks)
- (b) (i) Describe how igneous rocks are formed. (5 marks)
- (ii) List **three** examples of mechanically formed sedimentary rocks.(3 marks)
- (c) Explain **three** types of rock metamorphism. (6 marks)
- (c) Suppose you were to carry out a field study on rocks in the area surrounding your school
- (i) State **three** characteristics that you would look for while identifying different rock types (3marks)
- (ii) State **three** follow up activities for the field study. (3marks)

8. (a) (i) Differentiate between faulting and folding. (2 marks)
- (ii) Name **five** features formed as a result of faulting. (5 marks)

- (b) The world map below shows the location of some fold mountains. Use it to answer question (a) (i)



- (i) Identify the fold mountains marked W, X, Y and Z  
(4marks)
- (ii) Citing relevant examples, explain how Fold Mountains were formed according to the plate tectonics theory. (6 marks)
- (c) Explain the effects of Fold Mountains on the following
- (i) Transport. (2 marks)
- (ii) Agriculture. (2 marks)
- (iii) Tourism. (2 marks)
- (iv) Mining. (2 marks)
9. (a) Give **three** examples of West coast deserts. (3 marks)
- (b) (i) Apart from rock pedestals, name four other features formed due to wind erosion in arid regions. (4 marks)
- (ii) With the aid of well labelled diagrams, describe how a rock pedestal is formed. (7 marks)
- (c) Explain **four** factors that influence the rate of wind transport. (8 marks)
- (d) State **three** negative influence of desert features. (3 marks)
10. (a) (i) What is a lake? (2marks)
- (ii) Give **three** processes that lead to formation of Lakes. (3marks)

- (b) Describe how Lake Victoria was formed  
(6marks)
- (c) Explain **three** factors why some Lakes in the Rift Valley in Kenya are saline.(6marks)
- (d) Explain **four** ways in which lakes influence the climate of the surrounding areas.  
(8marks)

# FORM 3 MIDTERM 3 EXAMS

NAME: ..... ADM NO: ..... CLASS: .....  
312/2

## GEOGRAPHY

### PAPER 2

TIME: 2 ¾ HOURS

#### INSTRUCTIONS TO CANDIDATES

- ❖ This paper has two sections: A and B.
- ❖ Answer all the questions in section B.
- ❖ Answer question 6 and any other two questions from section B.
- ❖ Answer ALL questions on the provided foolscaps.

#### For Examiner's Use only.

| Section | Questions | Max. Score | Candidate's Score |
|---------|-----------|------------|-------------------|
| A       | 1 – 5     | 25         |                   |
| B       | 6         | 25         |                   |
|         | 7         | 25         |                   |
|         | 8         | 25         |                   |
|         | 9         | 25         |                   |
|         | 10        | 25         |                   |
|         | TOTAL     | 100        |                   |

#### SECTION :

Answer ALL the questions in this section.

1. (a) Name **TWO** places where diamond mining is done in the Republic of South Africa. (2 mks)  
(b) State **THREE** problems facing gold mining in South Africa. (3 mks)
2. (a) Name **TWO** major wheat growing counties in Kenya. (2 mks)

**FOR MARKING SCHEMES CALL/TEXT/WHATSAPP 0705525657**

- (b) Give **THREE** uses of wheat. (3 mks)
3. State **FIVE** characteristics of shifting cultivation. (5 mks)
4. Identify the characteristics of nomadic pastoralism. (5 mks)
5. (a) Name **THREE** ways in which placer mining can be done. (3 mks)
- (b) State **TWO** problems facing Trona mining on Lake Magadi. (2 mks)

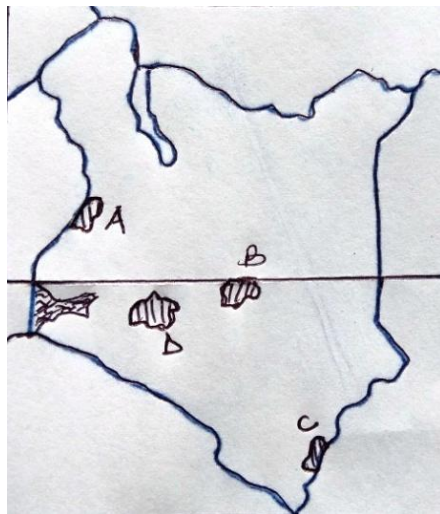
**SECTION:**

**Answer question 6 and any other two questions from this section.**

6. The table below shows the estimated exports of fruits from Kenya to the European market in '000 of tones between 2000-2004. Use it to answer the questions that follow.

| <b>Crop</b>  | <b>2000</b> | <b>2001</b> | <b>2002</b> | <b>2003</b> | <b>2004</b> |
|--------------|-------------|-------------|-------------|-------------|-------------|
| Oranges      | 50          | 45          | 42          | 38          | 30          |
| Apples       | 20          | 25          | 32          | 28          | 25          |
| Tangerines   | 32          | 35          | 40          | 38          | 35          |
| Melons       | 22          | 24          | 36          | 30          | 34          |
| <b>Total</b> | <b>124</b>  | <b>129</b>  | <b>150</b>  | <b>134</b>  | <b>124</b>  |

- (a) (i) Use a cumulative bar graph to present the data. (Use 1cm to represent 20 tonnes) (10 mks)
- (ii) Give two disadvantages of using cumulative bar graphs in data presentation. (2 mks)
- (b) Name five areas in Kenya where horticulture is practiced. (5 mks)
- (c) Explain four reasons why horticultural produce is exported by air to the European markets. (8 mks)
7. (a) State four characteristics of coniferous forests which favour their exploitation. (4 mks)
- (b) Explain five factors that favour the growth of natural forests on the slopes of Mt. Kenya. (10 mks)
- (c) Use the map of Kenya below to answer the following questions.



- (i) Name the forest reserves marked A, B, C and D. (4marks)
- (ii) Name **TWO** hardwoods commonly exploited in the forest reserve marked A. (2 mks)
- (d) State **FIVE** problems facing forestry in Kenya. (5 mks)
8. (a) (i) What is agriculture? (2 mks)
- (ii) Identify **FOUR** characteristics of plantation farming in Kenya. (4 mks)
- (b) Give **FOUR** physical factors that influence the growing of tea in Kenya. (4 mks)
- (c) Explain **FOUR** problems facing small scale tea farming in Kenya. (8 mks)
- (d) You plan to carry out a field study on a tea farm.
- (i) Give **FIVE** reasons why you need a route map. (5mks)
- (ii) Give **TWO** methods the class may use to gather data. (2 mks)
9. (a) Name **FOUR** sugarcane growing areas in Kenya. (4 mks)
- (b) Give **FOUR** human factors that influence successful sugarcane production (4 mks)
- (c) (i) Describe the stages involved in the processing of sugar cane at the factory. (6 mks)



- (ii) Give **THREE** by-products obtained from sugarcane. (3 mks)
- (d) Your class visited a sugarcane plantation for a field study on land preparation and planting.
- (i) Outline **FOUR** activities that you are likely to identify. (4 mks)
- (ii) Give **FOUR** methods you would use to record data during the study. (4 mks)
10. (a) (i) Define mining. (2 mks)
- (ii) Clearly state **FOUR** formations in which minerals occur. (2 mks)
- (b) (i) Name **THREE** methods of underground mining apart from shaft method.(3 mks)
- (ii) Describe how shaft method is carried out. (6 mks)
- (c) Explain **THREE** benefits of the oil mining to the local community in Turkana. (6 mks)

# FORM 3 MIDTERM 3 EXAMS

311/1

## HISTORY AND GOVERNMENT

### PAPER 1

NAME: .....ADM NO: .....CLASS:

.....

TIME: 2<sup>1</sup>/<sub>2</sub> HOURS

#### SECTION A (25 MKS)

*Answer all questions in this section.*

1. Give **two** limitations of using anthropology as a source of information on History and Government. (2mks)

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2. Give **one** evidence to show the Chinese reached the East African Coast. (1mk)

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3. List **two** communities that comprise the highland Nilotes. (2mks)

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4. Give **two** factors which influenced Seyyied Said to develop agriculture in Zanzibar. (2mks)

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5. Give the main reason why the colonial government created reserves in Kenya. (1mk)

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**FOR MARKING SCHEMES CALL/TEXT/WHATSAPP 0705525657**

6. State **two** characteristics of independent churches in Kenya during the colonial period. (2mks)

.....  
.....  
.....

7. Identify **one** method used by trade unionists to demand for their rights during the colonial period. (1mk)

.....

8. Name the first person to be appointed as a prime minister in Kenya. (1mk)

.....

9. Give **two** functions of the principal secretary in Kenyan government. (2mks)

.....  
.....  
.....

10. State **two** reasons why Nabongo Mumias of Wanga collaborated with the British. (2mks)

.....  
.....  
.....

11. Give **one** reason why corruption is being discouraged in Kenya. (1mk)

.....  
.....

12. State **one** way in which the government of Kenya has addressed the problems of landlessness. (1mk)

.....  
.....

13. Identify **two** political events which threatened the stability of Kenya 1975 to 1978. (2mks)

.....  
.....

.....  
.....  
14. Give **one** reason why Africans who lived in towns formed social welfare organizations during the colonial period. (1mk)  
.....  
.....  
.....

.....  
15. What was the main ideological difference between KANU and KADU before independence in 1963? (2mks)  
.....  
.....  
.....  
.....

16. Name the President of the Supreme Court in Kenya. (1mk)  
.....

17. Give the main reason why KANU refused to form a government after the 1961 election. (1mk)  
.....  
.....

### **SECTION B (45 MARKS)**

*Answer any **three** questions from this section.*

18. a) Give **three** reasons which contributed to the development of trade between the Kenya Coast and the outside world by the 16<sup>th</sup> Century. (3mks)

b) What reasons led to the decline of the Portuguese rule along the Kenyan Coast during the 16<sup>th</sup> century. (12mks)

19. a) Give **five** reasons why Christian missionaries established mission stations in Kenya during colonial period. (5mks)

b) Describe five factors which undermined Christian missionaries' activities in Kenya during the 16<sup>th</sup> century. (10mks)

20. a) Identify **three** African women who participated in the MauMau Movement. (3mks)

b) Describe the role of African women in the struggle for Independence in Kenya. (12mks)

**FOR MARKING SCHEMES CALL/TEXT/WHATSAPP 0705525657**

21. a) State **five** common grievances of the political organizations in Kenya up to 1939. (5mks)  
b) Explain **five** roles played by women in the Mau Mau Uprising. (10mks)

**SECTION C (30MKS)**

*Answer any **two** questions from this section.*

22. a) State **five** special groups that monitor human rights. (5mks)  
b) Explain **five** principles of democracy. (10mks)
23. a) Why are parliamentary elections held regularly in Kenya? (5mks)  
b) Explain **five** functions of the Executive committee in a county government in Kenya. (10mks)
24. a) State **five** ways through which the constitution of Kenya promotes national unity. (5mks)  
b) Explain **five** factors which make it difficult for the correctional department in Kenya to work effectively. (10mks)

# FORM 3 MIDTERM 3 EXAMS

311/2

## HISTORY AND GOVERNMENT

### PAPER TWO

NAME: .....ADM NO: .....CLASS: ....

TIME: 2<sup>1</sup>/<sub>2</sub> HOURS

#### SECTION A (25 MKS)

*Answer all questions in this section.*

1. Give **one** contribution of archaeology to the study of History. (1mk)

.....  
.....

2. Identify **two** activities that influenced early man to set up permanent settlement.(2mks)

.....  
.....  
.....  
.....

3. Identify **one** area in Africa where agriculture began. (1mk)

.....  
.....

4. State **two** disadvantages of human transport. (2mks)

.....  
.....  
.....  
.....

5. State **two** uses of bronze in ancient Benin. (2mks)

.....  
.....  
.....  
.....

6. Give the main reason for the decline of Meroe. (1mk)

**FOR MARKING SCHEMES CALL/TEXT/WHATSAPP 0705525657**

7. State **two** advantages of mobile phone as a means of communication. (2mks)

8. Give **two** uses of steam power in Europe during the 19<sup>th</sup> Century. (2mks)

9. State **one** function of the Lukiko in the Buganda Kingdom in the 19<sup>th</sup> Century. (1mk)

10. Mention **two** methods used by European powers to acquire colonies in Africa. (2mks)

11. State **two** privileges enjoyed by the assimilated Africans in the four French communes in Senegal. (2mks)

12. State **two** causes of the Ndebele war of 1893. (2mks)

.....  
.....  
.....  
.....  
13. Name the military wing of the African National Congress in South Africa. (1mk)

.....  
.....  
.....  
14. Name the person who convened the Berlin Conference. (1mk)

.....  
15. Give **one** early urban centers in pre-colonial Africa. (1mk)

.....  
16. Name the missionary chosen to represent Africans in the legislative council under the terms of Devonshire white paper of 1923. (1mk)

.....  
17. Name the commission mandated in matters concerning Election in Kenya. (1mk)

.....  
**SECTION B**

*Answer any three questions from this section.*

18. a) State **five** factors which led to the development of early agriculture in Mesopotamia. (5mks)

b) Explain **five** effects of Agrarian Revolution in Europe. (10mks)

19. a) State **three** results of the inventions of the textile industry in Britain.(3mks)

b) Explain **six** factors that led to Industrialization in Japan. (12mks)

20. a) State **five** causes of the Maji Maji Rebellion in Tanganyika (1905 – 1907). (5mks)

b) Explain **five** effects of the Chimurenga war of 1896 – 1897. (10mks)

21. a) State **three** roles of Kwame Nkurumah in the struggle for independence in Ghana. (3mks)

b) Explain **six** challenges faced by FRELIMO in the struggle for independence in Mozambique. (12mks)



**SECTION C (30 MAKS)**

*Answer any two questions from this section*

22. a) Outline **five** factors which contributed to the growth of Buganda Kingdom in the 19<sup>th</sup> Century. (5mks)  
b) Describe the social organization of the Asante Kingdom. (10mks)
23. a) Name **three** leaders charged alongside Nelson Mandela in the Rivonia trial of 1964. (3mks)  
b) Explain problems encountered by Nationalists in South Africa. (12mks)
24. a) State three privileges which were enjoyed by assimilated Africans in the four communes of Senegal. (3mks)  
b) Explain six results of the French policy of assimilation in Senegal. (12mks)

# FORM 3 MIDTERM 3 EXAMS

## KISWAHILI

JINA \_\_\_\_\_ NAMBARI \_\_\_\_\_ DARASA \_\_\_\_\_

102/1

### KISWAHILI KIDATO CHA TATU KARATASI YA 1

**1. Lazima**

Wewe uulifuzu kutoka chuo kikuu miaka mine iliyopita na shahada ya ualimu. Andika barua ya kuomba kazi ya ualimu na uiambatanishe na wasifukazi.

2. Usalama umekuwa changamoto nchini kwa muda sasa. Pendekeza hatua ambazo serikali yafaa kuchukua ili kuimarisha usalama.

3. Mstahimilivu hula mbivu

4. Andika insha itakayomalizika kwa maneno haya .... Niliyakumbuka maneno ya mwalimu mkuu, machozi yalinitoka njia mbilimbili. Hayawi hayawi huwa.

# FORM 3 MIDTERM 3 EXAMS

JINA \_\_\_\_\_ DARASA \_\_\_\_\_ NAMBAR  
I \_\_\_\_\_

102/2

KISWAHILI KIDATO CHA TATU

KARATASI YA PILI

LUGHA

MUDA: SAA 2½

## MAAGIZO

1. Jibu maswali yote.
2. Andika majibu yako katika nafasi ulizoachiwa katika kijitabu hiki cha maswali.
3. Karatasi hii ina kurasa **kumi**. Watahiniwa ni lazima waangalie kama kurasa zote za karatasi hii zimepigwa chapa sawasawa na kuwa maswali yote yamo.

## 1. UFAHAMU (ALAMA 15)

*Soma taarifa ifuatayo kisha ujibu maswali yanayofuata.*

Kwa kipindi cha miezi kadha iliyopita tumeshuhudia vyombo vya dola vikitia makali yake kwenye upekuzi na hata kupiga doria usiku na mchana katika jitihada za kulinda nchi.

Mpango huu ni kweli umeonekana kufanya kazi hivi kwamba hata magaidi wenyewe wameshindwa kupenya katika miji na sasa kuhiari kijinga kushambulia magari ya abiria, kitendo ambacho ni cha kuonyesha uwoga. Kwa hatua hiyo, navipa vyombo vya dola kongole. Jambo ambalo lafaa kujulikana ni kwamba mikakati ya kulinda nchi haifai kuwa ni ya wakati mmoja tu, mbali inafaa kuwa ni zoezi la kila siku.

Magaidi nao huwa macho huku yakijua bayana kwamba wakati wa kulala kwa walinzi unapokuwepo, basi wanapata nafasi ya kututupia ‘viazi’ ukipenda grunedi.

Kama ilivyo kawaida katika mataifa mengi barani Afrika, ni bayana kwamba bado kungali na mianya mingi ambayo magaidi wa kimataifa huendelea kutumia. Dosari bado zipo. Kwa mfano, mipaka mingi ya nchi hizi huwa kama lango kuu la ugaidi wa kimataifa, kwani kuenea kwa saratani ya ufisadi halimo tu maofisini mbali pia kwenye mipaka yetu.

Kama kupata kitambulisho, pasipoti na stakabadhi zingine za kusafiri nchini Kenya ashakum si matusi ziligeuzwa ‘maandazi’ ya Kariakoo basi niambie ni nani hawezi kuingia na kutoka nchini bila usumbufu wowote ule bora tu anayehitaji ana hela mkononi? Kwa kuikubali hongo kuwa ufunguo wa kila kitu, Wakenya wenzangu hapo naona ni kama

tumejiweka kwenye kikaango kilicho juu ya moto mkali. Hapa hakuna aliye na bahati, tajiri kwa masikini wamo kwenye mtego huu hatari.

Kwa mtindo ambao tunafuata wa kutoa ajira katika idara mbalimbali za ulinzi, inabidi serikali iwe na uangalifu sana hasa kwenye suala nzima la kuhakikisha stakabadhi wanazohitaji si ghushi.

Pasina kufanya hivyo hapo tena tunaweza kuwapata maadui wanaopenya na kujifanya walinzi wetu kumbe ni majasusi wa magaidi. Kila Mkenya anafaa kujihisi kulindwa. Miji, vijiji na hata vitongoji vinafaa kuwa na usalama wa kutosha, kwani kila Mkenya ni mlipa ushuru na hatufai kuona labda tabaka la juu likipendelewa huku mitaa ya mabanda ikiachiwa mbwa koko kama walinzi wao.

Suala lingine muhimu ni kuangaziwa upya usalama kwenye magari ya usafiri.

Juzi tulishuhudia mabasi mawili yakilipuliwa kwenye barabara ya Thika huku tukijua fika kwamba, mpango wa walinda nchi ungalipo.

La kusikitisha ni kuona kwamba, madereva na utingo wao walikamatwa na kufunguliwa mashtaka kwa kutozuia shambulizo hilo. Je, hii ni sheria gani? Dereva ataendesha gari au atachukua jukumu la walinda usalama?

Waswahili walinena kwamba ukubwa ni jaa na kwa hivyo Rais wa taifa ndiye anayefaa kubeba mzigo mzima wa usalama wetu bila kubananga wasaa. Wengine ambao wanafaa kuwajibika ni wakuu wote wa idara mbalimbali za usalama.

Usalama wako na wangu ni muhimu, elewa bayana kwamba bila usalama watalii hawawezi kuja kututembelea. Bila ya usalama maendeleo ya taifa kamwe hayawezi kupatikana, ndiposa kila jitihada sharti zifanywe ili wote waweze kuendelea kuyafurahia matunda ya uhuru wetu.

Ada ya mja kunena muungwana ni vitendo, hatufai tu kuimba wimbo wa ‘linda nchi’ ilhali mabasi barabarani hayana usalama. Hebu na tuone ulinzi endelevu na hapo tutawakomoa magaidi kwa yakini.

### **Maswali**

1. Ipe taarifa hii anwani mwafaka.

(alama 1)

---

2. Kwa nini magaidi wanashambulia magari ya abiria?

(alama 1)

3. Kulingana na taarifa pamoja na maoni yako, eleza njia tatu ambazo zinatumiwa na magaidi ili kufanikisha utekelezaji wa unyama wao.

(alama 3)

4. Ni njia gani ambazo magaidi hutumia kuingia katika nchi wanapoazimia kutekeleza uhalifu?

(alama 3)

5. Kwa nini suala la usalama kwenye magari ya usafiri muhimu?

(alama 1)

6. Ni njia zipi zinazoweza kutumiwa kupunguza mashambulizi ya kigaidi kulingana na mwandishi?

(alama 2)

7. Eleza athari za utovu wa usalama.

(alama 2)

8. Eleza neno au mafungu ya maneno kama yalivyotumiwa katika kifungu.

(alama 2)

- a) Vyombo vya dola

- b) Jaa

2. **UFUPISHO (ALAMA 15)**

***Soma taarifa ifuatayo kisha ujibu maswali.***

Ponografia ni tendo, maandishi, picha au mchoro unaoonyesha au kueleza uchi wa mtu au vitendo vya ngono kwa ajili ya kuchochea ashiki ya kuifanya. Mambo haya machafu huwasilishwa ana kwa ana kupitia sinema, video, magazeti, vitabu, muziki, televisheni. DVD, n.k.

Ponorafia imekuwepo tangu jadi, hasa katika nchi za magharibi. Lakini sasa limekuwa tatizo sugu. Hii ni kwa sababu imeenea ulimwenguni kote mithili ya moto katika mbuga wakati wa kiangazi. Uenezi umechangiwa na mambo kadha wa kadha. Mchango mkubwa zaidi umetokana na kuimarika kwa vyombo vya teknolojia ya habari na mawasiliano. Matumizi ya tarakilishi, mdahilishi na viungambali vya picha yamesambaza ponografia ilivyoanza. Hata hivyo, hubuniwa au kutengenezwa na makundi mbali mbali ya watu. Miongoni mwa hawa ni watu wasiojali maadili. Pili, kuna wale wenye matatizo ya kisaikolojia na kijamii. Wao hutengeneza na kueneza uchafu huu kwa lengo la ama kuvuruga maadili katika jamii au

kuchukiza wanajamii waadilifu. Kundi lingine ni lile la wanaoichukulia ponografia kama nyenzo ya kutosheleza ashiki zao. Hivi sasa, kundi kubwa ni lile la wanotumia matusi haya kama njia ya kuchuma. Kwa mfano, wanamziki ambao hutumia ponografia kuvutia wateja na hivyo kuzidisha mauzo yao.

Kushamiri kwa wimbi na uonyeshaji ponografia kuna athari kubwa kwa jamii na hasa watoto. Ingawa watu wengine hudai picha hizi haziwaathiri, upo ushahidi kuonyesha kuwa wanaotazama picha za ngono hupata matatizo. Lazima ieleweke kuwa kinachoonekana na jicho au kusikika kwa sikio huathiri fikira au hisia. Picha za matusi zinachangia kwa kiasi kikubwa kuharibu akili. Badala ya kuzingatia mambo muhimu kama masomo watu huanza kutafakari mambo machafu.

Vijana wengi ni kama bendera. Hivyo basi huanza kuiga wanayoyaona na kusikia. Hili ni tatizo linalowafanya kuacha mkondo wa maadili. Kutokana na uchafu huu, watu wengi hushawishiwa kuingilia shughuli za ngono mapema kabla hawajakomaa kimwili, kiakili na kihisia. Matokeo yake ni mengi. Haya ni pamoja na ukahaba, utendaji mbaya shuleni, mahudhurio mabaya darasani na mimba zisizotarajiwa. Vijana wengi huacha shule kabisa. Wengine nao huambukizwa magonjwa ya zinaa ambayo huwaletea mauti.

Inasemekana kuwa akili za binadamu hunata zaidi mambo yanayowasilishwa kwa picha. Si ajabu vijana huyadumisha matusi haya katika kumbukumbu zao na kuyasanya sehemu ya maisha yao. Wengi huanza kuandama tabia mbovu kama ushoga, ubasha na usagaji. Kuna wale ambao huanza kujichua. Kujichua ni hali ya mwanamke au mwanaume kumaliza haja za kimaumbile bila kufanya mapenzi na mtu mwingine. Ponografia imechangia pakubwa kuenea kwa haya.

Jambo hili limegeuza mielekeo ya vijana. Wanaiga mitindo mibaya ya mavazi yanayoanika uchi wao. Hali kadhalika, huiga lugha, ishara na miondoko inayohusiana na ngono. Yote haya yanapingana na desturi za Mwafrika. Si ajabu visa vya ubakaji vinaongezeka kila kukuchapo.

Utazamaji wa picha chafu aghalabu huandamana na maovu mengine kama unywaji pombe, matumizi ya dawa za kulevya, uvutaji sigara na utumiaji wa dawa za kuchochea uchu wa ngono. Mambo haya huwapa vijana kutazama tabia za unyama.

Jambo hatari ni kuwa kuendelea kutazama picha hizi huzifanya nishai na hisia za watu kuwa butu, yaani huondoa makali. Hata katika utu uzima, mtu atapoteza mhemko wa kawaida na kugeuzwa kuwa mtegemezi wa ponografia. Tatizo hili linaenea kwa vishindo mijini na vijijini. Ipo haja ya dharura kuikinga jamii kutokana na maenezi haya yasio na kizuizi.

Jambo la kwanza ni kuongeza ufahamu wa umma wote kuhusu uovu wa picha hizi. Kwa namna hii itawezekana kupunguza mahitaji na uuzaji wa ponografia. Tatizo hili linaenea kwa





- b) Kwa kutumia maneno 55 – 60, eleza mambo muhimu yanayojitokeza katika aya ya nne hadi ya tisa kuhusu athari za ponografia . (alama 9, 1 ya mtiririko)

**Matayarisho**

This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There are approximately 20 lines visible. The paper has a slight shadow on the right side, suggesting it's resting on a surface. There is no handwriting or other markings on the paper.

**FOR MARKING SCHEMES CALL/TEXT/WHATSAPP 0705525657**

1. Andika sifa bainifu za /d/ na /f/.

(alama 2)

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2. Andika kinyume cha sentensi hii.

(alama 2)

Mama alianika nguo zilizofuliwa na Kitwana.

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3. Andika neno 'kuja' katika hali ya kuamuru.

(alama 1)

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4. Weka shadda katika neno 'nta'.

(alama 1)

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5. Yakinisha sentensi ifuatayo.

(alama 2)

Wanafunzi wasipopita mtihani huo mwalimu hataumiwa.

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6. Kwa kutolea mifano, eleza miundo mitatu ya silabi za Kiswahili.

(alama 3)

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7. **Andika kwa udogo.**

(alama 2)

Chaka la Samba halilali paka.

8. **Changanua sentensi ifuatayo kwa kutumia jedwali.**

(alama 4)

Kiongozi aliyeng'olewa mamlakani amekufa.

9. **Mzizi ni nini? Toa mfano.**

(alama 2)

10. **Bainisha aina za virai vilivyopigiwa mstari.**

(alama 2)

Mjadala huo mzuri uliisha usiku wa manane.

11. **Tofautisha sentensi hizi:**

(alama 2)

- a) Ningeondoka sasa ningefika mapema.
- b) Ningaliondoka sasa ningalifika mapema.

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12. **Toa maana mbili za sentensi hii.**

(alama 2)

Aliletewa ng'ombe na mtoto wake.

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13. **Tunga sentensi moja kubainisha.**

(alama 2)

Kivumishi cha pekee cha kusisitiza.

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14. **Nyambua vitenzi vifuatavyo katika kauli zilizo kwenye mabano.**

(alama 2)

- a) la (fanyia)

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- b) suka (tendata)

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15. **Fafanua aina ya hali zilizotumika katika sentensi hizi.**

(alama 3)

- a) Mumo akachekeka, akafurahia na akalala.

b) Mtoto wa Maria hulia kila mara.

c) Mwalimu aandika ubaoni.

16. Tumia neno 'karibu' katika sentensi kama:

(alama 2)

a) Kihusishi

b) Kihisishi

17. Tenganisha mofimu katika neno lifuatalo kisha uonyeshe majukumu yake kisarufi.

(alama 3)

Onana

18. Bainisha yambwa katika sentensi hii.

(alama 2)

Simu aliyonunuliwa Ali na mjombake imepotea.

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19. Tumia kivumishi kionyeshi cha mbali pamoja na nomino katika ngeli ya I–I.

(alama 1)

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4. ISIMU JAMII (ALAMA 10)

a) Eleza wajibu wa lugha ya Kiswahili kama lugha ya taifa nchini Kenya.

(alama 5)

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b) Fafanua sifa za sajili ya matanga.

(alama 5)



# FORM 3 MIDTERM 3 EXAMS

## KISWAHILI

Karatasi ya 3

FASIHI

Muda: Saa 2 ½

KIDATO CHA TATU

### Maagizo

- (a) Jibu maswali **manne** pekee.
- (b) Swali la **kwanza** ni la **lazima**.
- (c) Maswali hayo mengine **matatu** yachaguliwe kutoka sehemu **tatu** zilizobaki..
- (d) Usijibu maswali **mawili** kutoka **sehemu moja**.
- (e) Majibu yote **lazima** yaandikwe kwa lugha ya Kiswahili.
- (f) **Karatasi hii ina kurasa 7 zilizopigwa chapa.**
- (g) **Watahiniwa ni lazima wahakikishe kwamba kurasa zote za karatasi hii zimepigwa chapa sawasawa na kuwa maswali yote yamo.**

### SEHEMU A: RIWAYA

CHOZI LA HERI( ASSUMPTA K. MATEI)

1. “Wewe ni mfuata mvua...”
  - a) Eleza muktadha wa dondoo hili. ( alama 4)
  - b) Taja na ufafanue sifa tano za msemazwa. ( alama 10)
  - c) Je, ukabila umeathiri vipi jamii inayosawiriwa katika Riwaya hii? ( alama 6)

### SEHEMU B: TAMTHILIA

**FOR MARKING SCHEMES CALL/TEXT/WHATSAPP 0705525657**

**TAMTHILIA YA KIGOGO (PAULINE KEA)**

2. “Ukitaka kula asali,kaa na nyuki.”

a) Eleza muktadha wa dondoo hili. ( alama 4)

b) Bainisha mbinu ilivyotumika katika dondoo hili. (Alama 2)

c)Eleza jinsi wahusika wafuatao walivyofaidika kwa kukaa na nyuki. (Alama 14)

i) Chopi

ii) Kenga

iii) Ngurumo

iv) Mamapima

**au**

3. Tamthilia ya Kigogo ni kioo cha uhalisia wa maisha ya jamii nyingi za Kiafrika. Thibitisha.

**SEHEMU D: USHAIRI**

**4 . SHAIRI LA KWANZA**

**Soma shairi hili kisha ujibu maswali.**

Jukwani naingia, huku hapa pasokota,

Kwa uchungu ninalia,hii tumbo nitaikata,

Msiba mejiletea,nimekila kiso takata,

We tumbo nitakupani,uwe umetosheka?

Wazee hata vijana,wote umewasubua,

Huruma nao hauna,heshima kawakosea,

Ukambani na Sagana,hata mbwa wararua,

We tumbo nitakupani,uwe umetosheka?

Wahasibu ofisini,kibwebwe mejifunga,  
Miaka mingi vitabuni,ili wasikose unga,  
Nadhari wanadhamini,hesabu wanazirenga,  
We tumbo nitakupani, uwe umetosheka?

Wapenzi wa kiholela,pia wanakuogopa,  
Baada yao kulala, wana wao wanatupa,  
Wakihitaji chakula,wanachokora mapipa,  
We tumbo nitakupani,uwe umetosheka?

Wafugaji hata nao,kama dawa wakwamini,  
Hawajali jiranio,wamesusia amani,  
Wanaiba ng'ombe wao,na kuzua kisirani,  
We tumbo nitakupani,uwe umetosheka?

Nayo mizozo ya maji, kaonekana kwa mara,  
Hiyo nayo ni dibaji,sababu sio harara,  
Njaa wahepe wenyeji, huo ndio mkarara,  
We tumbo nitakupani,uwe umetosheka?

Ningeweza kukuuza,ingekuwa siku njema,  
Tena kwa bei ya meza,sokoni nimesimama,  
Wala tena singewaza,kuhusu wali na sima,  
We tumbo nitakupani,uwe umetosheka?

Hatima umefikika, naenda zangu nikale,  
Mate yanidondoka, kwa mnukio wa wale,  
Naomba kwenda kukaa, wala sio nikalale,  
We tumbo nitakupani, uwe umetosheka?

**Maswali**

- i Lipe anwani mwafaka shairi hili. (Alama 2)
  - ii Shairi hili ni la aina gani? Toa sababu. (Alama 2)
  - iii Huku ukitolea mifano mwafaka, taja arudhi zilizotumiwa katika ubeti wa tatu. (Alama 4)
  - iv Andika ubeti wa nne kwa lugha nathari. (Alama 4)
  - v Thibitisha kuwepo kwa idhini ya ushairi. (Alama 2)
  - vi Taja madhila anayoelezea mtunzi wa shairi hili yaletwayo na tumbo. (Alama 4)
  - vii Elezea maana ya maneno yafuatayo. (Alama 2)
- (a) Dibaji
- (b) Harara

**au**

**5. SHAIRI LA PILI**

**Soma shairi lifuatalo kwa makini kisha ujibu maswali.**

Niokoa Muokozi, uniondolee mashaka.  
Kuyatukua siwezi, mjayo nime dhikika  
Nime kithiri simanzi, ni katika kuudhika  
Mja wako nasumbuka, nipate ni yatakayo.

Mja wako nasumbuka, nataka kwao afua  
Nirehemu kwa haraka, nami nipate pumua

Naomba hisikitika, na mikono hiinua

Mtenda ndiwe Moliwa, nipate niyatakayo.

Mtenda ndiwe Moliwa, we ndiwe Mola wa anga

Mazito kuyaondoa, pamoja na kuyatenga

Ukauepusha ukiwa, ya pingu zilonifunga

Nikundulia muwanga, nipate niyatakayo

Muwanga nikundulia, nipate toka kizani

Na huzuni n'ondolea, itoke mwangu moyoni

Mambo mema niegheshea, maovu nisitamani.

Nitendea we Manani, nipate niyatakayo.

Igeuze yangu nia, dhaifu unipe mema

Nili katika dunia, kwa afia na uzima

Moliwa nitimizia, yatimize yawe mema

Nifurahike mtima, nipate niyatakayo.

(a) Shairi hili ni la bahari gani? Eleza.

(alama 2)

(b) Taja madhumuni ya shairi hili.

(alama 3)

(c) Eleza muundo wa shairi hili.

(alama 4)

(d) Thibitisha namna uhuru wa kishairi unaibuka katika shairi.

(alama 4)

(e) Andika ubeti wa pili katika lugha sufufu.

(alama 4)

(f) Toa maana ya:

(i) Nimedhikika

(ii) Muwanga nikundulia

(iii) Nifurahike mtima

(alama 3)

**SEHEMU D: FASIHI SIMULIZI**

**6.**

Ewe kilizi

Ulozowe kujiificha

Nyuma ya mama kujikinga, dhidi ya milio

Ya radi lo juu mbinguni

Jua kesho ni siku ya siku

Siku ya kujua mbichi na mbivu

Kutofautisha jogoo na vipora,

ngariba taposhika , chake kijembe

Ndipo utakapojua bayani

Ukoo wetu si wa kunguru

Ikiwa hu tayari

Kisu kikidhihaki

Sithubutu kamwe, wanjani kuingia

Sije kuniaibisha miye, amiyo na akraba nzima!

a) Nani anaimba wimbo huu? Thibitisha. ( alama 2)

b) Wimbo huu ni wa aina gani? ( alama 2)

c) fafaua sifa tano za nyimbo za aina hii. ( alama 5)

d) Nyimbo za aina hii zina majukumu gani katika jamii? ( alama 5)

e) Thibitisha mbinu tatu za kifasihi zilizotumiwa katika utungo huu. ( alama 6)

**au**

7. “Dhima ya pekee ya hadithi ni kuburudisha.” Jadili. ( alama 20)

# FORM 3 MIDTERM 3 EXAMS

NAME.....ADM  
NO.....CLASS.....

## MATHS

### PAPER 1

TIME: 2 ½ HOURS

SECTION I-(answer **all** the 16 question in this section-50MARKS)

1. Use tables of square roots and reciprocals tables to evaluate to 3 decimal places the problem below.

$$\frac{10}{\sqrt{0.625}} + \frac{4}{\sqrt{164}}$$

(3marks)

2. The heights of two similar pails are 12cm and 8cm. The larger pail can hold 2 litres. What is the capacity of the smaller pail?  
(3 marks)

3. Find the equation of the perpendicular bisector of the line AB where A is (3, 9) and B is (7, 5) giving your answer in the form  $ax + by + c = 0$ .  
(3 marks)



4. Mr. Ochuodho who deals in electronics sells a radio to a customer at Kshs. 1,440 after giving him a discount of 10% but finds that he still makes a 20% profit. Find the profit Mr. Ochuodho would make if he does not give a discount.

(3mks)

5. Simplify the expression

$$\frac{9t^2 - 25a^2}{6t^2 + 19at + 15a^2}$$

(3marks)

6. Solve for x in  $\left(\frac{4}{9}\right)^x \times (8)^{1-x} = 486$

(4marks)

7. The length and width of a rectangle are stated as 18.5cm and 12.4cm respectively. Both measurements are given to the nearest 0.1cm.

- a) Determine the lower and upper limit of each measurement.  
(1 mark)
  
- b) Calculate the percentage error in the area of the rectangle.  
(3 marks)

8. In a regular polygon, each interior angle is  $x^\circ$  and each exterior angle is  $\left(\frac{x-36}{3}\right)^\circ$

- (i) Find angle  $X^\circ$   
(1mk)
  
- (ii) Find the number of sides of the polygon  
(2mks)

9. Find the integral values of  $x$  which satisfy the following inequalities;

$$2x + 3 > 5x - 3 > -8$$

(3mks)

10. Two boys and a girl shared some money. The younger boy got  $\frac{5}{18}$  of it; the elder boy got  $\frac{7}{12}$  of the remainder and the girl got the rest. Find the percentage share of the younger boy to the girl's share.  
(3mks)

11. Simplify the expression below leaving your answer in rationalized surd form of  $a + b\sqrt{c}$

$$\frac{1 + \tan 120^\circ}{1 + \cos 330^\circ}$$

(3mks)

12. Mutua bought 8 pairs of trousers and six shirts at Sh. 4160. If he had bought twice as many shirts and half as many trousers, he would have saved Sh. 160. Find the cost of each item  
(3 marks)

13. A solid block in the shape of a cylinder has a height of 14cm and weighs 22kg. If it is made of material of density  $5\text{g/cm}^3$ , find the radius of the cylinder. Take  $\pi = \frac{22}{7}$

(3mks)

14. The number 5.81 contains an integral part and a recurring decimal. Convert the number into an improper fraction and hence a mixed fraction.

(3 Marks)

15. Using a pair of compasses and a ruler only construct a triangle ABC such that AB = 4cm, BC = 6cm and angle ABC =  $135^\circ$ .

(3mks)

16. The curved surface area of a cylindrical container is  $1980\text{cm}^2$ . If the radius of the container is 21cm, calculate to one decimal place the capacity of the container in litres

(3 mks)

(Take  $\pi = \frac{22}{7}$ ).

**SECTION II ANSWER ANY FIVE QUESTIONS**

17(a) Train A leaves a station 45 minutes before train B. Both trains travel in the same direction and their speeds are 36km/h and 48km/h respectively.

i) How long will it take train B to catch up with train A? (3 marks)

ii) How far from the start were the two trains when they met. (2 marks)

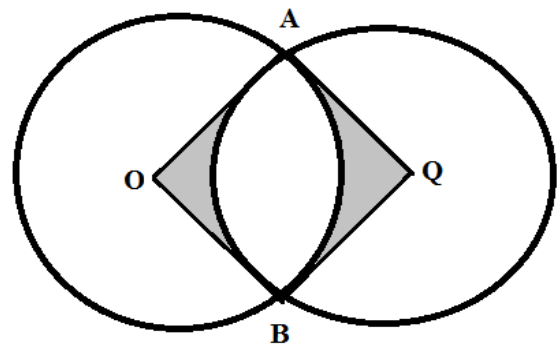
b) A car accelerated from rest to a velocity of 10m/s in 10 seconds. It travelled at this velocity for 20 seconds and then came to a stop in 5 seconds. Find;

i) The initial acceleration. (2 marks)

ii) The distance travelled. (2 marks)

iii) The average velocity. (1 mark)

18. Two circles with centres O and Q and radii 8cm intersect at points A and B as shown



below.

Given that the distance between O and Q is 12cm and that the line AB meets OQ at X, find:

(a) the length of the chord AB. (3marks)

(b) the reflex angle AOB. (3marks)

(c) the area of the shaded region.  $\pi = 3.142$

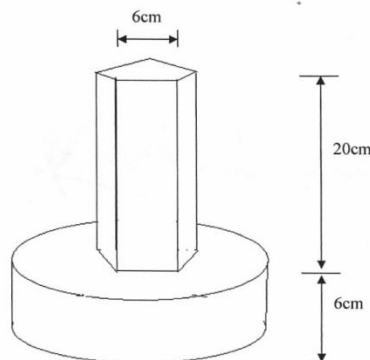
(4marks)

19. A triangle whose vertices are A(1,4) B(2,1) and C(5,2) is given the following transformations.

- i) A reflection along the line  $y = x$  to  $A^1B^1C^1$
- ii)  $A^1B^1C^1$  is given a rotation of a positive quarter turn about the origin  $A^{11}B^{11}C^{11}$
- iii)  $A^{11}B^{11}C^{11}$  is given an enlargement of linear scale factor -2 about (1,2) to  $A^{111}B^{111}C^{111}$

- a) Using the grid provided, plot the triangle ABC and its image  $A^1B^1C^1$  (3mks)
- b) Locate the image  $A^{11}B^{11}C^{11}$  from the grid hence state its co-ordinates. (3mks)
- c) Find the co-ordinates of  $A^{111}B^{111}C^{111}$  hence plot it on the grid (4mks)

20. The figure below shows a prism whose cross section is a regular pentagon of side 6cm and whose length is 20cm  
14cm and height 6cm to  
joined to a cylinder of radius 14cm and height 6cm to form the model of a solid



(a) Calculate the cross section area of the pentagon  
(3mks)

(b) Calculate the total volume of the solid  
(4mks)

(c) The model represents a pillar of total height 5.2m, calculate the volume of the actual solid in  $\text{m}^3$   
(3mks)

21. An expedition has 5 sections AB, BC, CD, DE and EA. B is 200m on a bearing of  $050^\circ$  from A. C is 500m from B. The bearing of B from C is  $300^\circ$ . D is 400m on a bearing  $230^\circ$  from C. E is 250m on a bearing  $025^\circ$  from D.

(a) Sketch the route  
(1 Mark)

- (b) Use the scale of 1cm to 50m to draw the accurate diagram representing the route.  
(5 Marks)

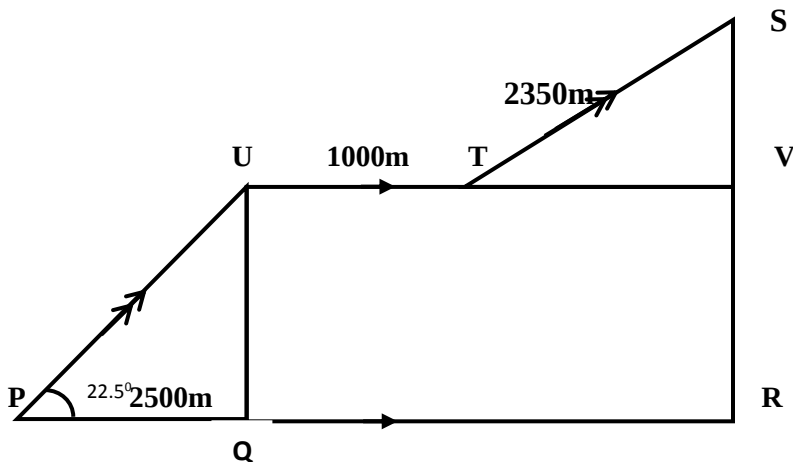
**SKETCH**

**TO SCALE**

- (c) Use your diagram to determine  
(i) Distance in metres of A from E  
(2 Marks)

- (ii) Bearing of E from A  
(2MKS)

22. In the figure below  $PQ = 2500\text{m}$ ,  $UT = 1000\text{m}$  and  $TS = 2350\text{m}$ .  $PQR$  is a straight line. Parallel to  $UT$  and angle  $UPQ = 22.5^\circ$ .





Calculate to the nearest meter

(a) (i) U Q  
(2marks)

(ii) T V  
(2marks)

(iii) V S  
(2marks)

(iv) P U  
(2marks)

(b) Find the perimeter of the figure.  
(2marks)

23. A circular lawn is surrounded by a path of uniform width of 7m. The area of the path is 21% that of the lawn.

(a) Calculate the radius of the lawn.  
(4 mks)

- (b) Given further that the path surrounding the lawn is fenced on both sides by barbed wire on posts at intervals of 10 metres and 11 metres on the inner and outer sides respectively.  
Calculate the total number of posts required for the fence.  
(4 mks)

- (c) Calculate the total cost of the posts if one post costs sh 105.  
(2 mks)

24. A surveyor recorded the measurements of a field book using  $XY=400\text{m}$  as the base line as shown below.

|          |     |          |
|----------|-----|----------|
|          | Y   |          |
| To E 200 | 320 |          |
| 150      | 210 | To D     |
| To F 250 | 170 | 150 To C |
|          | 50  | 225 To B |
|          | X   | 100 To A |

- a) Use a scale of 1cm to represent 50m to draw the map of the field. (5mks)

- b) Find the area of the field in hectares (5mks) (*more working space at back page*)

# FORM 3 MIDTERM 3 EXAMS

NAME.....ADM

NO.....CLASS.....

## MATHEMATICS

### PAPER 2

TIME: 2½ HOURS

### SECTION I (50 MARKS)

*Answer ALL questions in the spaces provided.*

1. Evaluate using mathematical tables only. (3 marks)

$$\frac{6.373 \log 4.948}{\sqrt{0.004636}}$$

2. Determine the inverse of the matrix  $\begin{pmatrix} 4 & 3 \\ 5 & -2 \end{pmatrix}$ . Hence find the coordinates of the point at which the two lines  $4x - 18 = -3y$  and  $5x - 2y = 11$  intersect. (4 marks)

3. Solve the equation.

$$5^{2x+1} - 3(5^{x+1}) + 10 = 0$$

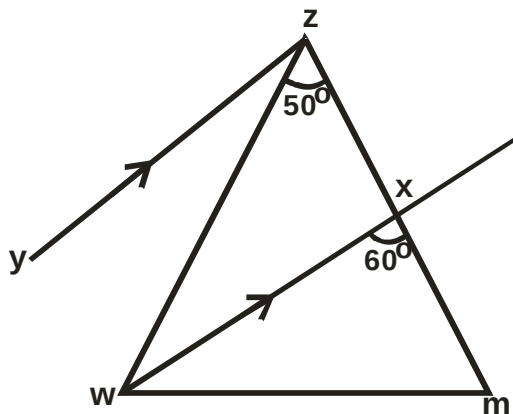
(4 marks)

4. Solve for  $x$  in the equation  $\cos x = \sin (3x - 30)$ . Hence determine the value of  $\tan x$  leaving your answer in surd form.

(3 marks)

5. In the figure below,  $YZ$  is parallel to  $WX$ . Angle  $WZX = 50^\circ$  and angle  $WXM = 60^\circ$ . Determine the size of angle  $YZW$ .

(2 marks)



6. Kaula has two types of coffee, costing Ksh.120 and Ksh.200 respectively. He mixed them in the ratio 6: 5 by weight.
- a) Determine to the nearest shilling the cost of one kilogramme of the mixture.
- (2 marks)

b) Find the percentage profit on the cost price if the mixture was sold at Ksh. 250.

(Give your answer in 2 d.p).

(2 marks)

7. Rationalise the denominator and simplify leaving your answer in the form  $\sqrt{a} + b$ .

(3 marks)

$$\frac{\sqrt{2} + 2\sqrt{5}}{\sqrt{5} - \sqrt{2}}$$

8. Use squares, square roots and reciprocals tables only to evaluate;

$$\frac{3}{\sqrt{42.15}} + \frac{4}{(3.152)^2}$$

(4 marks)

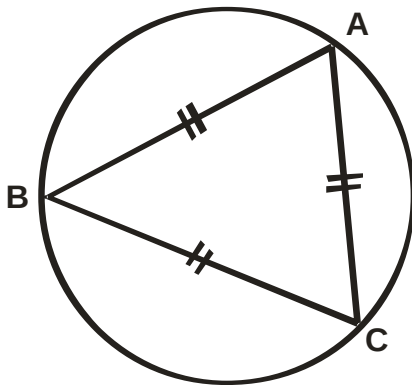
9. Make  $h$  the subject of the formula in

$$m = \frac{p}{\sqrt{h} + k^2}$$

(3 marks)

10. The diagram below shows an equilateral triangle ABC inscribed in a circle of radius 9cm. Calculate the length of the sides of the triangle (2 d.p)

(2 marks)



11. Agotho has a rectangular plot that was measured to the nearest meter and found to be 80m in length and 60m in width. Determine the percentage error in its perimeter.

(3 marks)

12. A circle of radius 3cm has its centre at  $(3, -2)$ . Express the equation of the circle in the form  $x^2 + y^2 + mx + ny + c = 0$ . Where  $m$ ,  $n$  and  $c$  are constants.  
(3 marks)

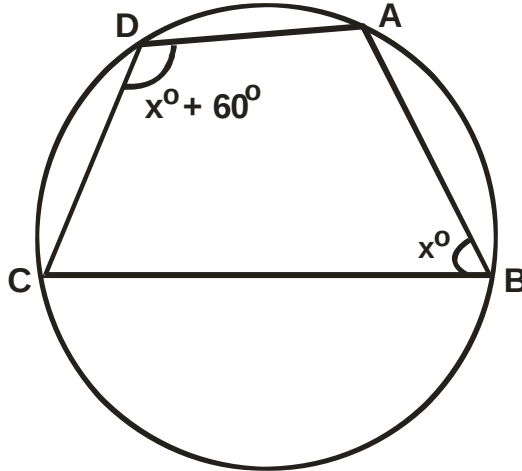
13. Wambua invested Sh. 6400 at 15% per annum compound interest for 3 years. Muinde invested twice that amount at  $12\frac{1}{2}\%$  per annum simple interest for the same period of time. Find whose investment earned more interest and by how much.  
(4 Marks)

14. a) Expand  $\left(1 - \frac{1}{2}x\right)^5$   
(1 mark)

- b) Use the expansion upto  $x^3$  in (a) above to evaluate  $(0.98)^5$  correct to 4 d.p  
(2 marks)



15. The figure below shows a quadrilateral ABCD which is cyclic. Solve for x.  
(2 marks)



16. Solve for x in the equation  $\log (x - 1) = \log 12 - \log (x - 2)$   
(3 Marks)

### **SECTION II (50 MARKS)**

Answer only **FIVE** questions from this section in the spaces provided.

17. Mr. Kobe is a civil servant who earns a monthly salary of Ksh.21200. He has a house allowance of Ksh.12000 per month, other taxable allowances are commuter Ksh.1100, medical allowance Ksh.2000. He is entitled to a personal relief of Ksh.1240 per month. Using the income rates below, solve the questions that follow.

| Income in Ksh. per month | Rates in Ksh per sh 20 |
|--------------------------|------------------------|
| 1 – 8,400                | 2                      |
| 8401 – 18,000            | 3                      |
| 18001 – 30,000           | 4                      |
| 30001 – 36,000           | 5                      |
| 36001 – 48,000           | 6                      |
| Above 48,000             | 7                      |

Determine;

- a) i) His monthly taxable income.  
(2 marks)

- ii) Net tax (PAYEE)  
(5 marks)

- b) In addition to the PAYEE, the following deductions were made. Ksh.250 for NHIF, Ksh.120 service charges, he repays a loan at sh.4500 and contributes towards savings at sh.1800 every month. Calculate his net salary per month.  
(3 marks)

18. a) From whole numbers 1 to 10, a number is selected at random, find the probability that, the number selected is a prime or a multiple of 3.

(2 marks)

b) A tetrahedron is thrown and a coin is tossed.

- i) List down all the possible events in the probability space.

(1 mark)

- ii) Find the probability of getting at least 2 and a head.

(2 marks)

- c) i) A bag contains 6 white marbles and some brown ones. If the probability of picking a brown marble is 0.6, find the number of marbles in the bag.

(3 marks)

ii) Two marbles are then picked, one at a time from the bag in c(i) above, with replacement.

Find the probability that the marbles picked are of different colours.

(2 marks)

**19.** Kennedy bought three cows and twenty-five goats spending a total of Sh. 75000. If he had bought two cows and thirty three goats, he would have saved Sh. 5400. Kennedy later sold all his animals at a profit of 40% per cow and 50% per goat.

Determine;

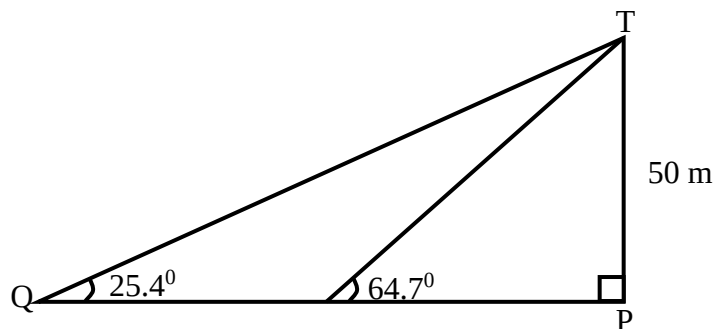
(a) the cost at which he bought each animal.

(5 Marks)

(b) the total amount of money Kennedy received after selling all the animals.

(5 Marks)

20. The figure below shows the position of a boat Q which is observed sailing directly towards the pier P at the base of a vertical cliff PT. The angle of elevation of the top of the cliff from Q is  $25.4^\circ$ . After 14 seconds the boat is at point R, and the angle for elevation of T is now  $64.7^\circ$ .



If the cliff is 50m high, calculate

- (a) The distance PQ

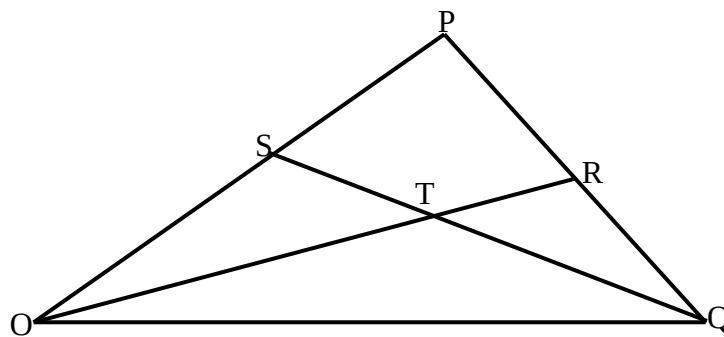
(2 Marks)

- (b) The distance QR

(4 Marks)

- (c) The speed of the boat in km/h  
(4 Marks)

21.



In the figure above,  $OPQ$  is a triangle in which  $QS = \frac{3}{4}QP$  and  $PR:RQ = 2:1$

Line  $OR$  and  $SQ$  meet at  $T$ .

- (a) Given that  $\vec{OP} = \vec{p}$  and  $\vec{OQ} = \vec{q}$ , express the following vectors in terms of  $\vec{p}$  and  $\vec{q}$ .

(i)  $\vec{PQ}$

(1 Mark)

(ii)  $\vec{OR}$

(2 Marks)

(iii)  $\vec{SQ}$

(1 Mark)

- (b) You are further given that  $ST = mSQ$  and  $OT = nOR$ . Determine the values of  $m$  and  $n$ .  
(6 marks)

22. Two variables  $A$  and  $B$  are connected by the equation.  $A = kB^n$  Where  $k$  and  $n$  are constants.

The table below gives values of  $A$  and  $B$ .

A. 1.5    1.95    2.51    3.20    4.50

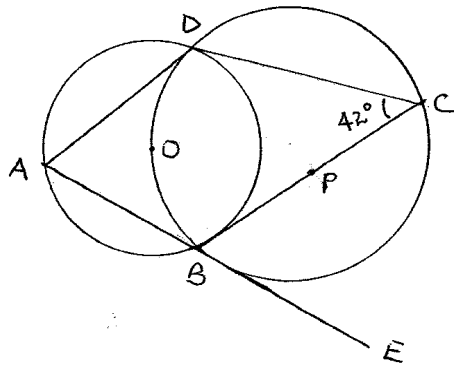
B. 1.59    2.51    3.98    6.31    11.5

- (a) Find a linear equation connecting  $A$  and  $B$   
(2 Marks)

- (b) On the graph paper provided draw a suitable straight line graph to represent the relation in (a) above (5 Marks)

- (c) Use your graph to estimate the values of  $k$  and  $n$  in to one decimal place.  
(3 Marks)

23. (a)



Giving reasons, determine the size of:

a) Angle CBD (2 marks)

b) Angle ODB (2 marks)

c) Angle BAD (2 marks)

d) Angle ABC (2 marks)

e) Angle ODA (2 marks)

24. (a) Draw the graph of the function below on the grid provided

$$y = 2x^2 - 7x - 2 \text{ for the values of } -1 \leq x \leq 6 \quad (5 \text{ marks})$$

(b) From your graph determine the roots of the function.  $2x^2 - 7x - 2 = 0$ . (1 marks)



- (c) By drawing a suitable graph of function  $y = 2x - 7$  on the same axis, solve the simultaneous equations  $y = 2x^2 - 7x - 2$  and  $y = 2x - 7$ . (4 marks)

# FORM 3 MIDTERM 3 EXAMS

NAME:.....CLASS:....ADM

NO:...

## PHYSICS

PAPER 1

2 HOURS

### SECTION A (25MARKS)

#### INSTRUCTIONS.

Answer all the questions in the spaces provided.

1. State the reading shown on the scale of a vernier calipers in the diagram below. (1mk)

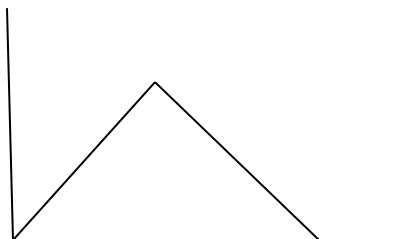


2. State the reason why electricity transmission cables are left sagging between the pylons. (1mk)

3. The reading on a mercury barometer at the foot of a hill. Density of air =  $1.25\text{kg/m}^3$ , density of mercury =  $1.36 \times 10^4\text{kg/m}^3$ . (3mks)

4. The figure below shows the displacement time graph for the motion of an object.

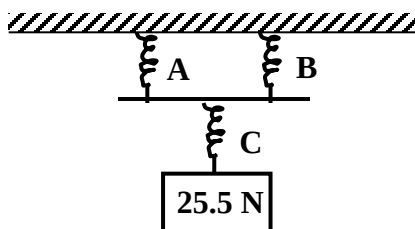
Displacement(m)



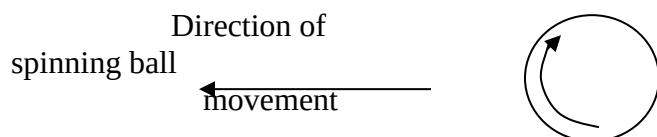
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- Time (s)
- Sketch the velocity time graph for the object. (2mks)
5. State two factors that would raise the boiling point of water. (2mks)
6. The level of water in a burette is  $25\text{cm}^3$ . 40 drops each of volume  $0.05\text{cm}^3$  are added to the burette, what would be its new reading. (2mks)
7. Explain how the efficiency of a vacuum flask is affected if the double-walled glass surface is replaced with a double-walled metal surface. (2mks)
8. A body moving at  $50\text{m/s}$  decelerates uniformly at  $2\text{m/s}^2$  until it comes to rest. What distance does it cover from the time it starts to decelerate to the time it comes to rest. (3mks)
9. Sketch a graph of pressure versus volume for an ideal gas at constant temperature. (3mks)

10. Three identical springs A, B, and C are used to support 25.5N weight as shown below. The weight of the horizontal bar is 2.5N, determine the extension on each spring given that 6N causes an extension of 2cm. (2mks)



11. State the property of Freon that makes it useful as a refrigerant liquid. (1mk)
12. The diagram below shows a spinning ball as it moves through air in the direction shown. (1mk)



Draw the streamlines of air around the ball and show the direction in which it spins such that an upward force is created. (2mks)

13. State the law of conservation of energy. (1mk)

### SECTION B (55 MARKS)

14. State one renewable source of energy. (1mk)

b. An electric crane lifts a load of 2000kg through a vertical distance of 3m in 6 seconds.

Determine:

- i. The work done. (2mks)

ii. The power developed by the crane. (2mks)

iii. The efficiency of the crane if operated by an electric meter rated 12.5kw. (2mks)

c. A pulley system has two pulleys on the lower block and one pulley on the upper block. Sketch the system showing the direction of the rope, the load and the effort. (3mks)

15. Define the term friction. (1mk)

b. A spherical steel ball is released from rest just above the surface of a column of oil which is in a long wide tube.

i.State three forces affecting the movement of the ball as it falls in the oil. (3mks)

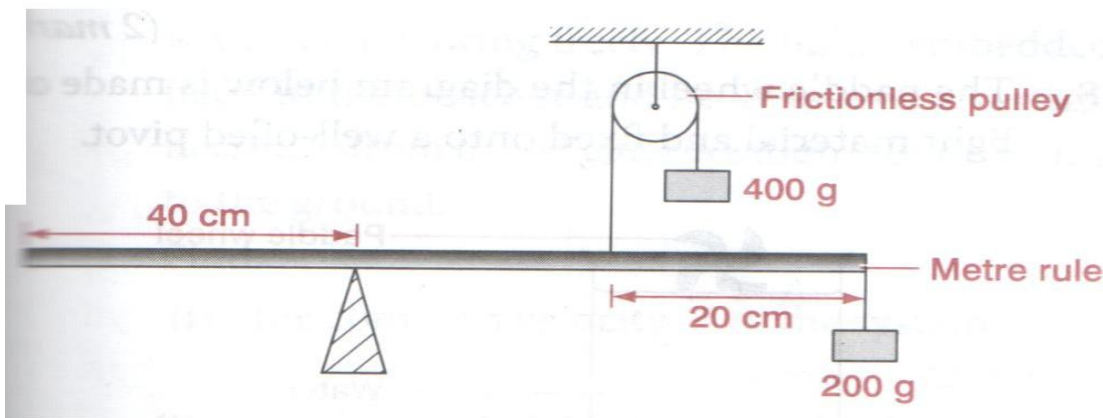
ii.Explain which of the forces in b (i) above varies as the ball falls. (1mk)

iii.What is the net force acting on the ball that is moving at terminal velocity?(1mk)

iv.Sketch a graph to show the variation of resultant force on the ball with velocity from the moment it was released. (2mks)

c. State the principle of movements. (1mk)

i. The diagram below shows a uniform metre rule in equilibrium under the forces shown.



ii. Determine the weight of the metre rule. (3mks)

16. A 4kg mass is receiving heat at a rate of 100KJ per minute and its temperature at various times recorded as follows.

|                    |     |     |     |     |     |     |     |     |     |     |     |
|--------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Temp<br>(°C)       | 230 | 250 | 270 | 270 | 270 | 270 | 310 | 350 | 390 | 390 | 390 |
| Time<br>t<br>(min) | 0   | 1   | 2   | 6   | 9   | 13  | 18  | 23  | 28  | 29  | 30  |

a. Plot a graph of temperature against time. (5mks)

b. Use your graph to determine:

i. Specific heat capacity of the substance in solid state. (3mks)

ii. Specific heat capacity in its liquid state. (3mks)

iii. Its melting point (1mk)

iv. Its boiling point (1mk)

v. Specific latent heat of fusion. (2mks)

17. In an experiment to demonstrate Brownian motion, smoke was put in a smoke cell and observed under a microscope.

i. State and explain the observation made. (2mks)

ii. Give a reason for using small particles like smoke particles in the experiment. (1mk)

iii. What observation would be made if the temperature of the smoke cell is raised? (1mk)

b. An oil drop of average diameter 0.7mm spreads out into a circular patch of diameter 75cm on the surface of water in a trough.

i. Calculate the thickness of the molecule. (3mks)

ii.State any two sources of error in b (i) above. (2mks)

18. At  $30^{\circ}\text{C}$  , the pressure of a gas is 100cmHg. At what temperature would the pressure of the gas fall by 20cm of mercury. Give the temperature in  $^{\circ}\text{C}$ . (3mks)

b. A hole of area  $4.0\text{cm}^2$  at the bottom of a tank 5m deep is closed with a cork. Determine the force on the cork when the tank is filled with water. ( $g=10\text{m/s}^2$ , density of water =  $1000\text{kg/m}^3$ ). (4mks)

c. A measuring cylinder of height 25cm is filled to a height at 15cm with water and the rest is occupied by kerosene. Determine the pressure acting on its base. (Density of water =  $1\text{g/cm}^3$ , density of kerosene =  $0.8\text{g/cm}^3$  atmospheric pressure = 103,000 Pa) (3mks)



# FORM 3 MIDTERM 3 EXAMS

NAME.....ADM.....CLASS  
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## 232/2 PHYSICS

### PAPER 2 (THEORY)

TIME: 2 HRS

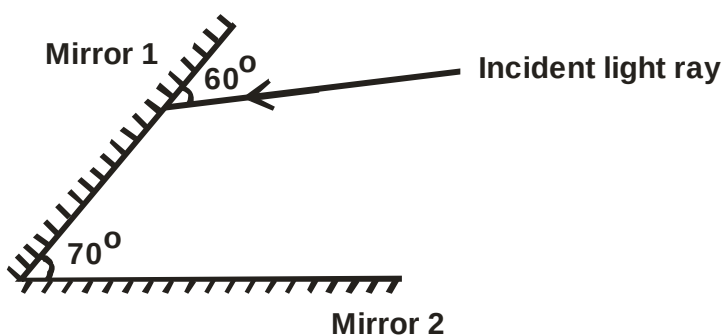
#### INSTRUCTIONS TO CANDIDATES

1. Mathematical tables and silent electronic calculators **may be** used.
2. The following constants may be used where necessary
  - Plancks constant,  $h = 6.63 \times 10^{-34} \text{ JS}$
  - Charge of an electron,  $e = 1.6 \times 10^{-19} \text{ C}$
  - 1 electron volt (1eV)  $= 1.6 \times 10^{-19} \text{ J}$

#### SECTION A (25 MARKS)

*Answer ALL the questions in this section in the spaces provided*

1. The figure below shows a ray of light incident on the surface of one plane mirror.



Sketch the path of the ray on the diagram after striking mirror 2 indicating all the angles. (2 marks)

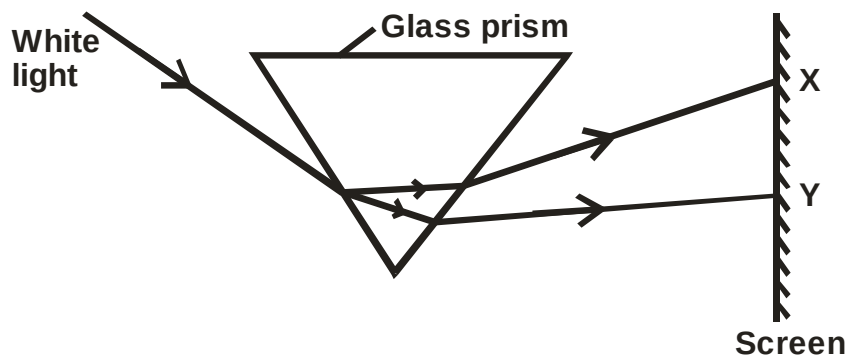
2. Explain why eight dry cells of 1.5V each arranged in series to give a total e.m.f of 12.0V cannot be used to start a car just like a lead-acid accumulator.

(2 marks)

3. When a candle flame is brought near the cap of a charged electroscope, the electroscope discharges. Explain this observation.

(2 marks)

4. The figure below shows an experiment set up used to illustrate dispersion of white light.



a) Identify the colours X and Y.

(1 mark)

X \_\_\_\_\_

Y \_\_\_\_\_

b) What physical property changes for the colours when they enter the prism?

(1 mark)

5. Diffraction is not a common phenomenon in light. Explain why.

(1 mark)

6. Apart from temperature, state any **one** other factor which affects the resistance of a metallic conductor.

(1 mark)

7. a) Define the term capacitor.

(1

mark)

b) Two plates of a parallel-plate capacitor are 0.6 mm apart and each has an area of 4 cm<sup>2</sup>. Given that the potential difference between the plates is 100 V, calculate the charge stored in the capacitor.

(Take ( $\epsilon_0 = 8.85 \times 10^{-12} \text{ Fm}^{-1}$ )

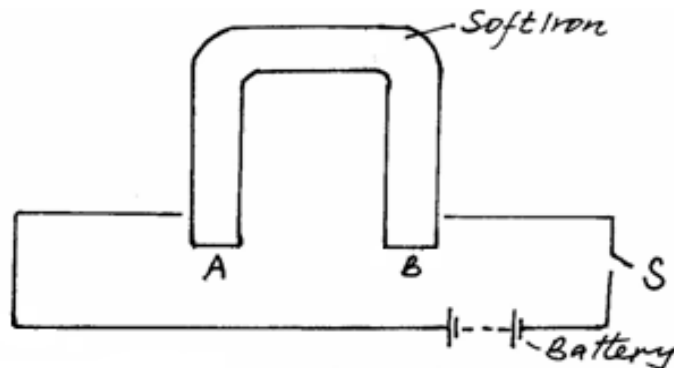
( 3mks)

8. A current of 0.8A flows through an electric circuit. Determine the quantity of charge that passes a point in the circuit in 6 minutes.

(2

marks)

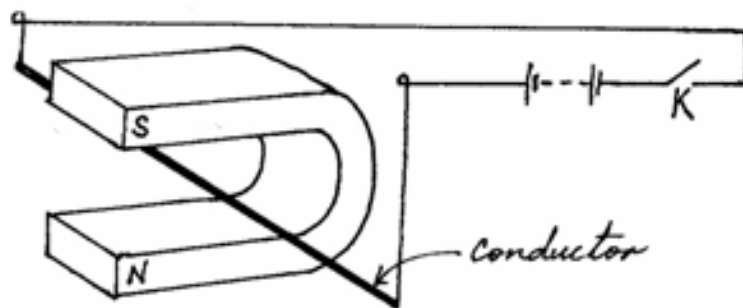
9. The figure below shows an incomplete circuit of an electromagnet.



Complete the circuit by drawing the windings on the two arms of the core so that A and B are both north poles when the switch S is closed. Indicate the direction of the current of the windings drawn. (1 mark)

10. A girl standing 120m away from a tall building claps her hands and hears an echo 0.75s later. Determine the speed of sound in air at this place. (3 marks)

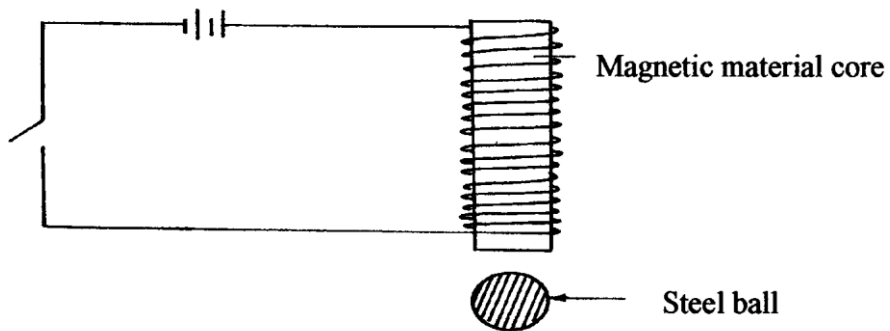
11. The figure below shows a thick copper conductor placed between two poles of a strong magnet. The wire is free to swing in between the poles.



(a) Indicate on the same diagram the direction in which the conductor swings when the switch K is closed. (1 mark)

(b) State one change that can be made on the set up so that the direction of swing of the conductor is reversed. (1 mark)

12. The set up in figure 4 below can be used in a laboratory for lifting and releasing a steel ball.



i) State the material which is suitable for use in the core. (1mark)

i) If a slightly larger ball is to be lifted, it is necessary to make an electromagnet stronger.

Name **two** ways of increasing the strength of the magnet.

(2 marks)

**SECTION B (55 MARKS)**

*Answer ALL the questions in this section in the spaces provided.*

13. a) Sound is a mechanical longitudinal wave. Explain why sound is classified as;

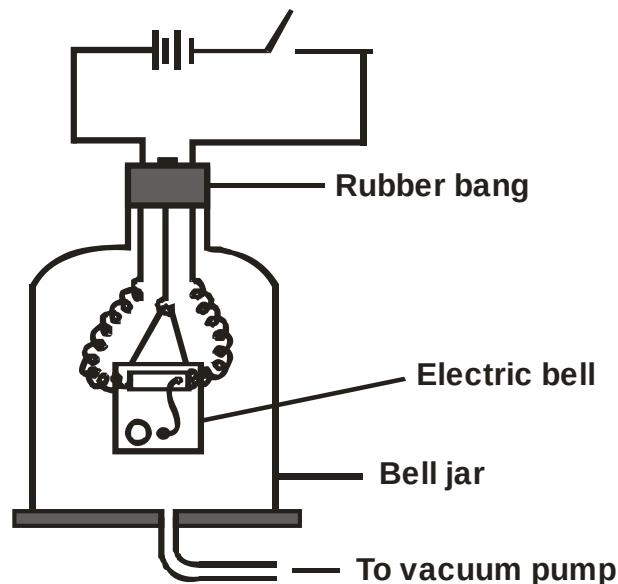
i) A mechanical wave.

(1 mark)

ii) A longitudinal wave.

(1 mark)

b) The diagram below shows a set up that was used to demonstrate that, sound is a mechanical wave.



i) State the function of the vacuum pump.  
(1 mark)

ii) When the switch is closed, state and explain what happens as the air is continuously drawn from the bell jar.  
(3 marks)

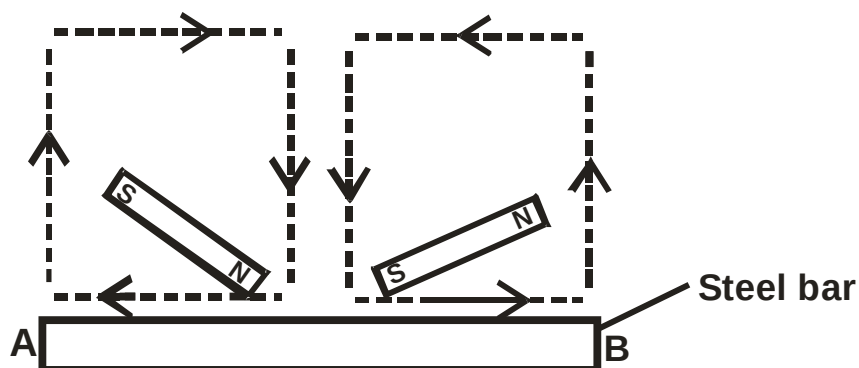
iii) State **two** possible reasons why it is not possible to reduce the sound completely in this experiment.  
(2 marks)

c) A fishing boat uses ultrasound of frequency  $6.0 \times 10^4 \text{ Hz}$  to detect fish directly below. Two echoes of the ultrasound are received, one after 0.09 seconds coming from the shoal of fish and the other after 0.12 seconds coming from the sea bed. The sea bed is 84m below the ultrasound transceiver.

i) State **two** reasons why ultrasound is preferred.  
(2 marks)

- ii) Calculate the speed of the ultrasound waves in water.  
(2 marks)

14. a) The figure below shows a method of magnetization used in making magnets.



- i) Name the method.  
(1 mark)

- ii) Identify the polarities A and B of the magnet produced.  
(2 marks)

A \_\_\_\_\_

B \_\_\_\_\_

- iii) Apart from this method, state any other method used in magnetization.  
(1 mark)

- b) In demagnetization by electrical method:



i) State the type of current used. (1 mark)

ii) Explain your answer in (i) above. (2 marks)

a) Explain why when demagnetizing a magnet, the magnet should be held in the East-West direction. (2mrks)

15. (a) Define the following terms ( 3 mks)

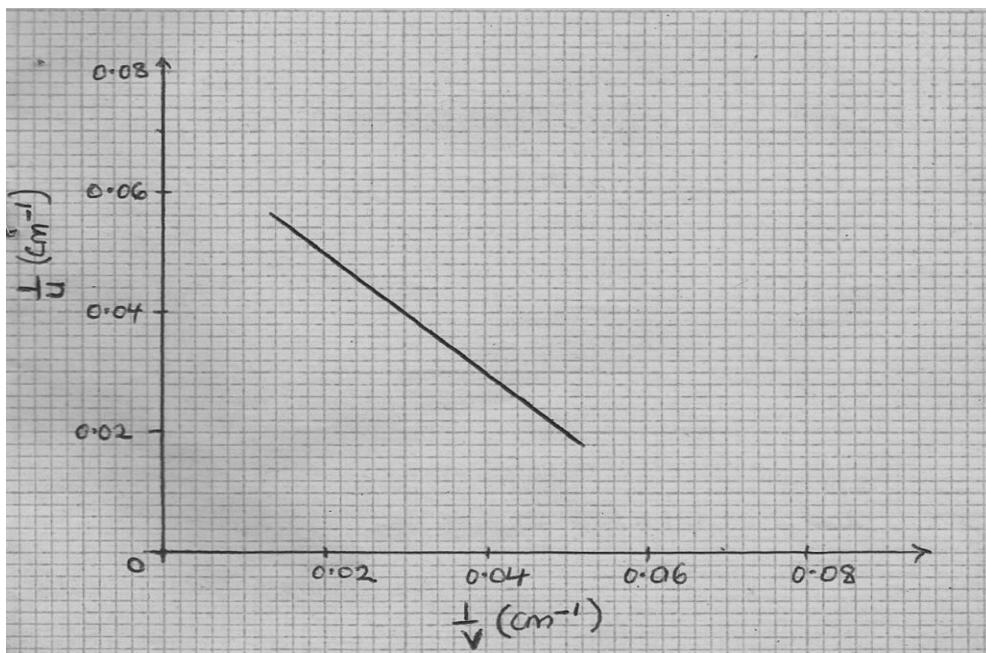
i) Principal focus

ii) Centre of curvature

iii) Focal plane

(b) ) An object is placed 18 cm in front of a concave mirror of focal length 12 cm. Determine the position and nature of the image formed (3mks)

- (c) The graph below shows variation of  $\frac{I}{u}$  ( $\text{cm}^{-1}$ ) with  $\frac{I}{v}$  ( $\text{cm}^{-1}$ ) for an object placed in front of a concave mirror

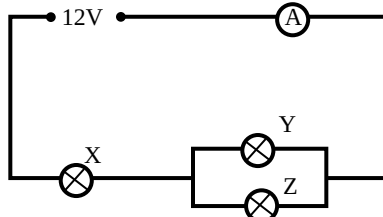


From the graph,

- (i) Determine the focal of the mirror. (2mks)

- (ii) Determine the image distance when the object is 20cm from the mirror. (2mks)

16. (a) The figure below shows how a student set up a circuit using 3 identical bulbs X, Y and Z each rated “12V, 2.0A”



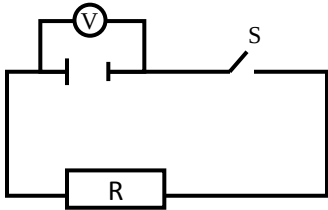
(i) When operating normally, calculate the resistance of one of the bulbs (2 Marks)

(ii) Calculate the effective resistance of the three bulbs. (2 Marks)

(iii) What will be reading of the ammeter? (2 Marks)

(iv) Draw a circuit diagram showing the three bulbs connected in such a way that they would all work at the same brightness especially if they are not identical. (2 Marks)

(b) When the switch S is kept open in the circuit shown below the voltmeter reads 1.5V. When the switch is closed, the readings drops to 21.3V and the current through the resistor is 0.5A.



(i) What is the e.m.f of the cell?  
(2 Marks)

(ii) What the terminal voltage of the cell?  
(1 Mark)

(iii) Calculate the value of R.  
(3 Marks)

17. a) Sketch a graph of displacement against time for a transverse wave of frequency of 50Hz  
of at least two cycles with amplitude 2cm. (4marks)

- b) Distinguish electromagnetic waves and mechanical waves  
(2marks)

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- c) A pulse-echo sounder is used by fishing boat to locate a shoal of fish in water. The sounder sends sound of frequency 21KHz and wavelength of 7.5cm. if the echo is received after 0.4seconds, determine how far the shoal of fish is from the base of the boat. (4marks)

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